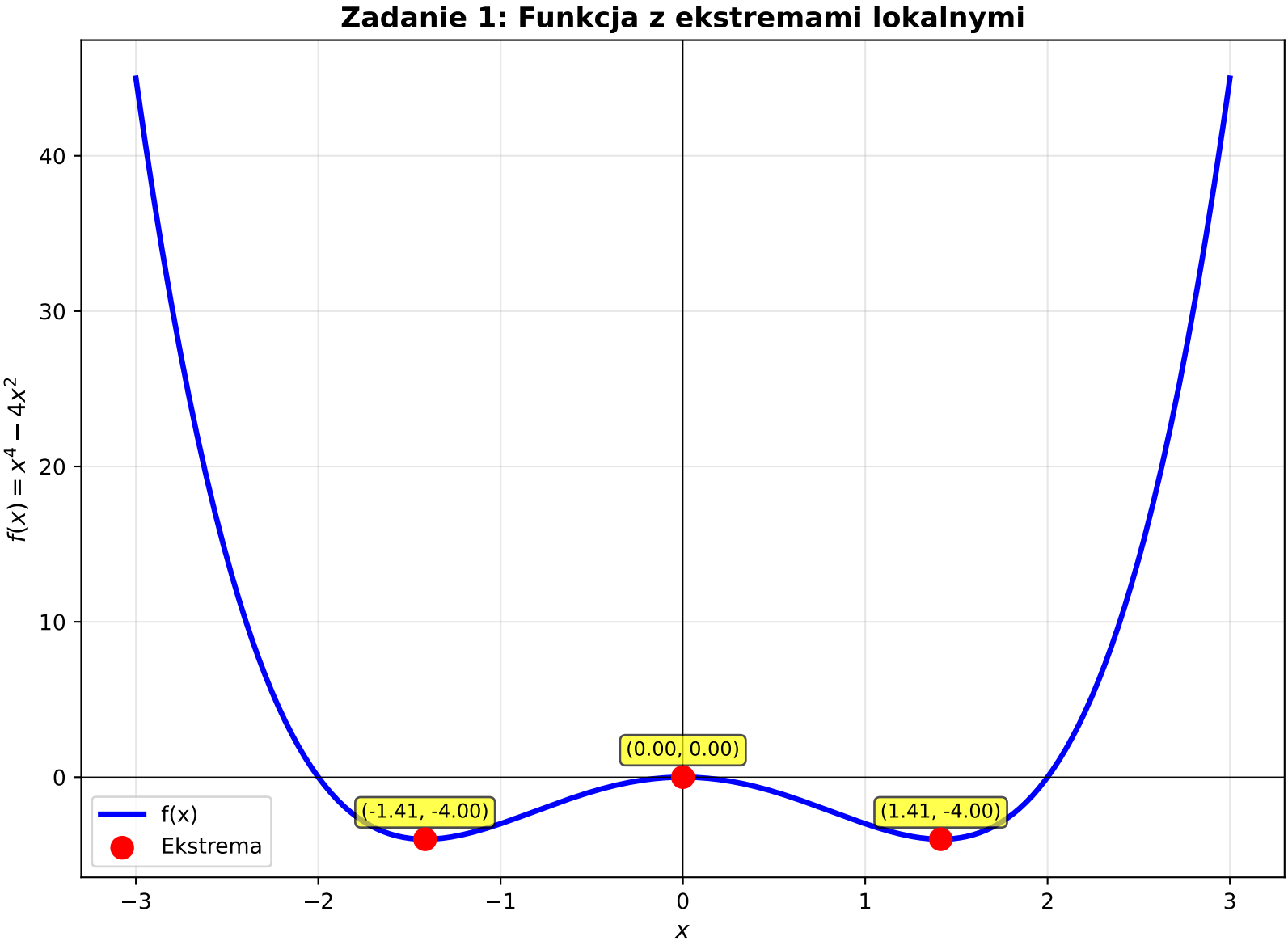
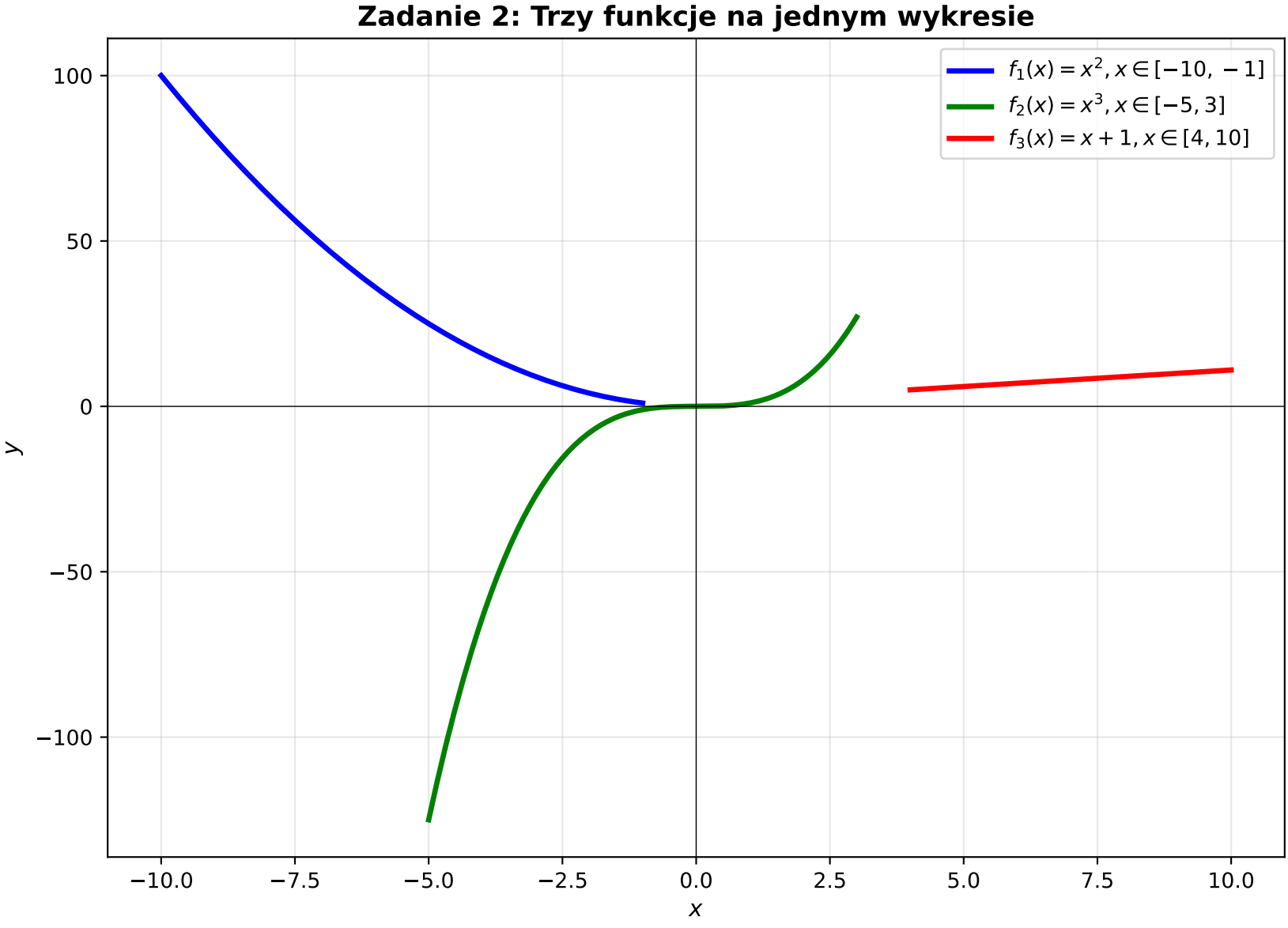
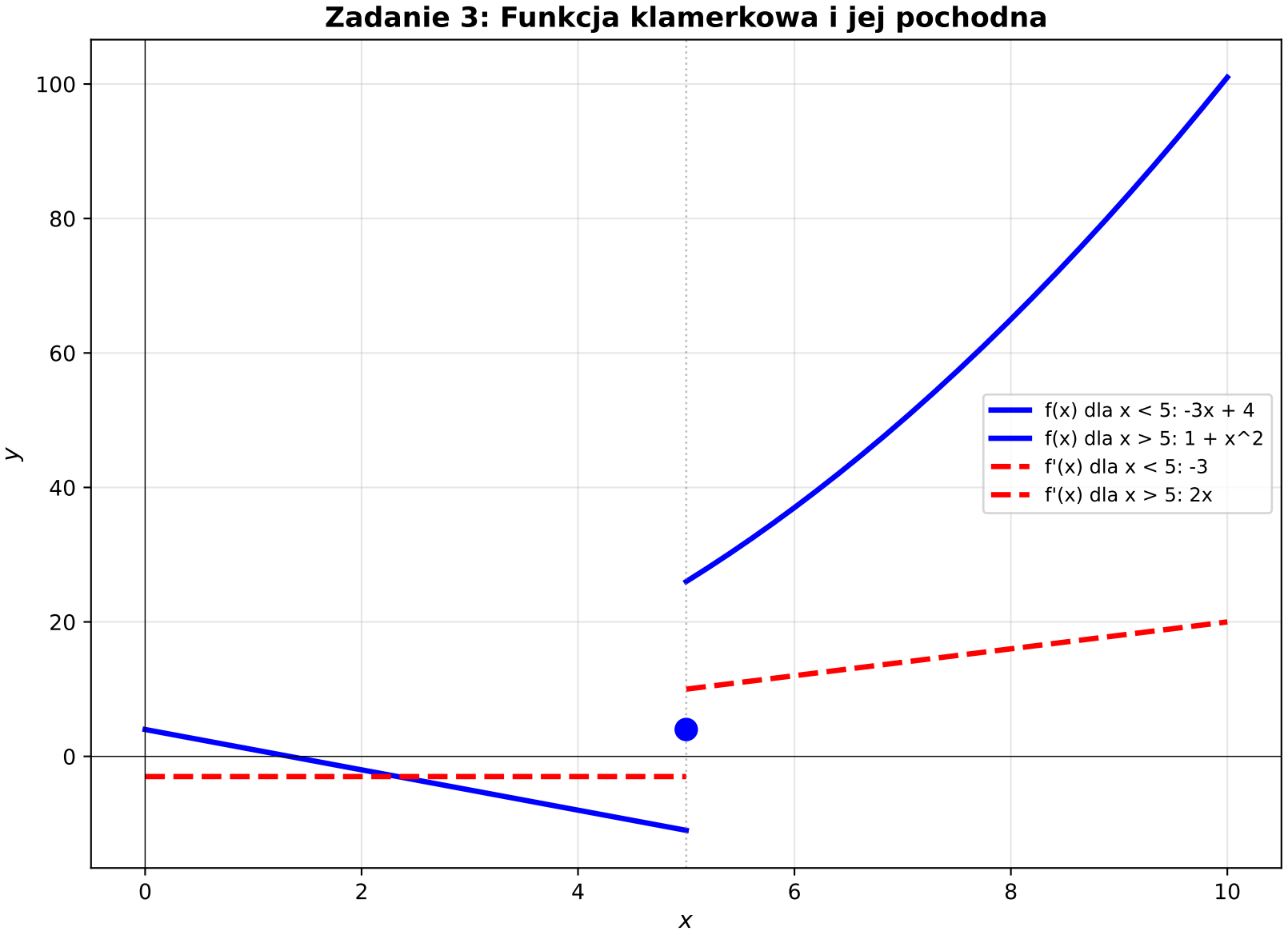
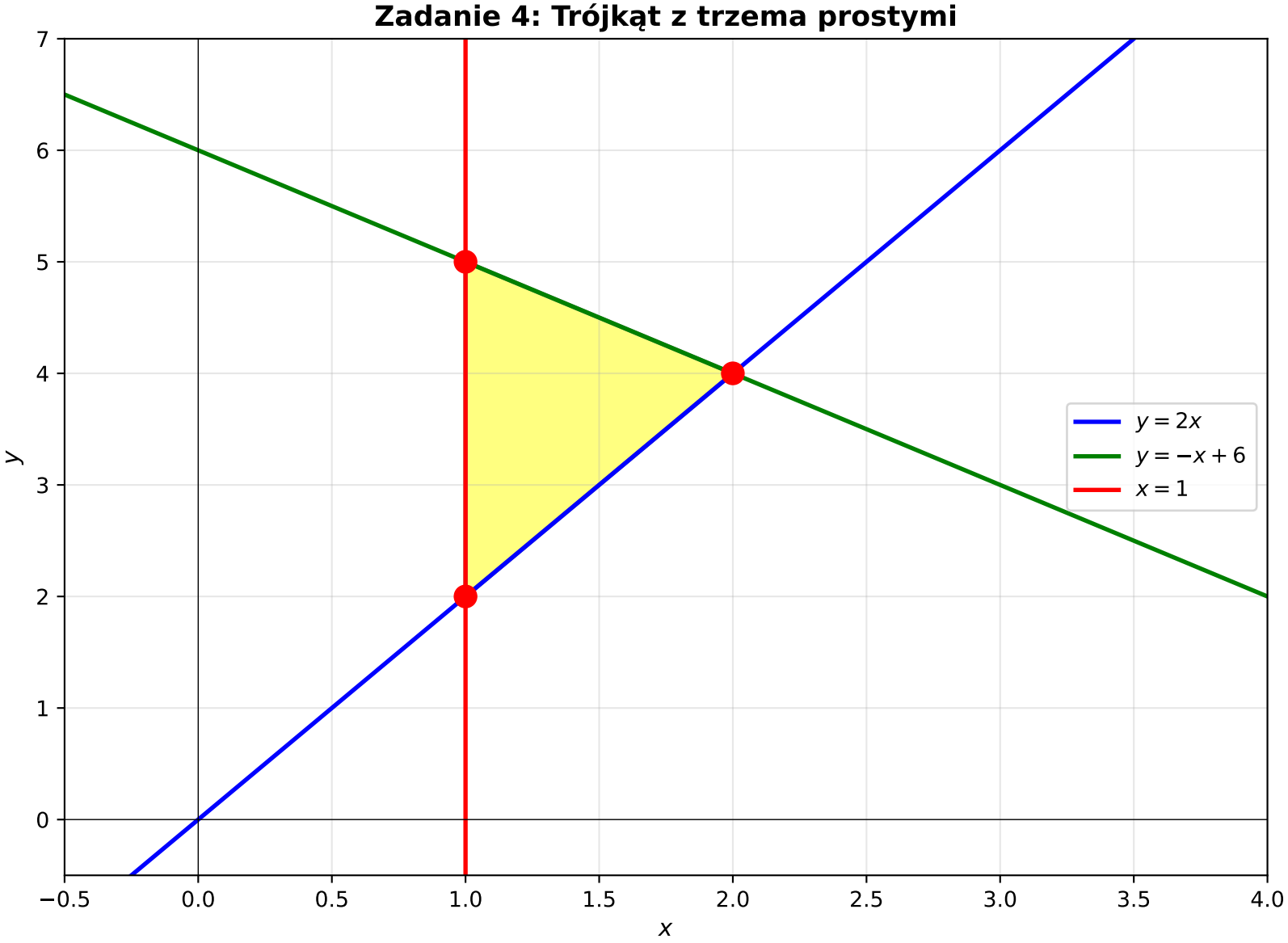
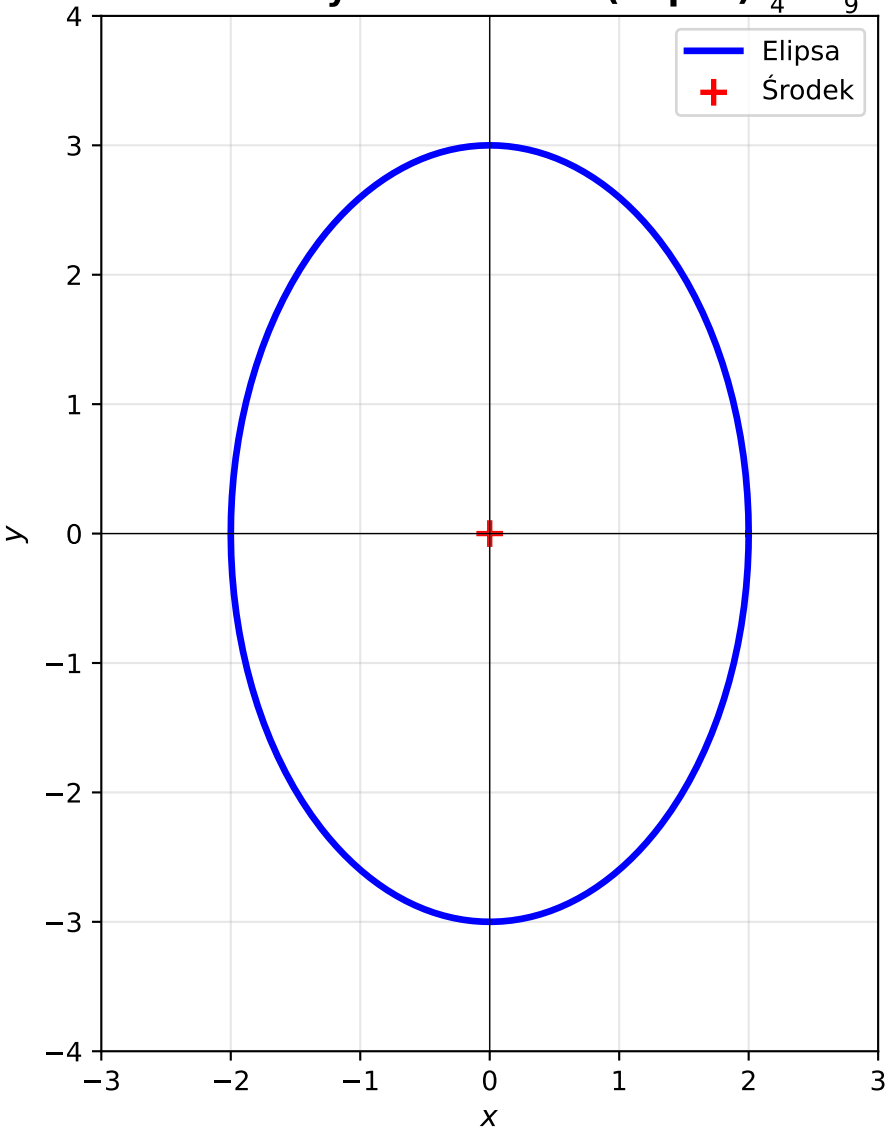

Zadanie 1: Funkcja z ekstremami lokalnymi
$f(x) = x^4 - 4x^2$
x
f(x)
Ekstrema
(0.00, 0.00)
(-1.41, -4.00)
(1.41, -4.00)
40
30
20
10
0
−3
−2
−1
0
1
2
3

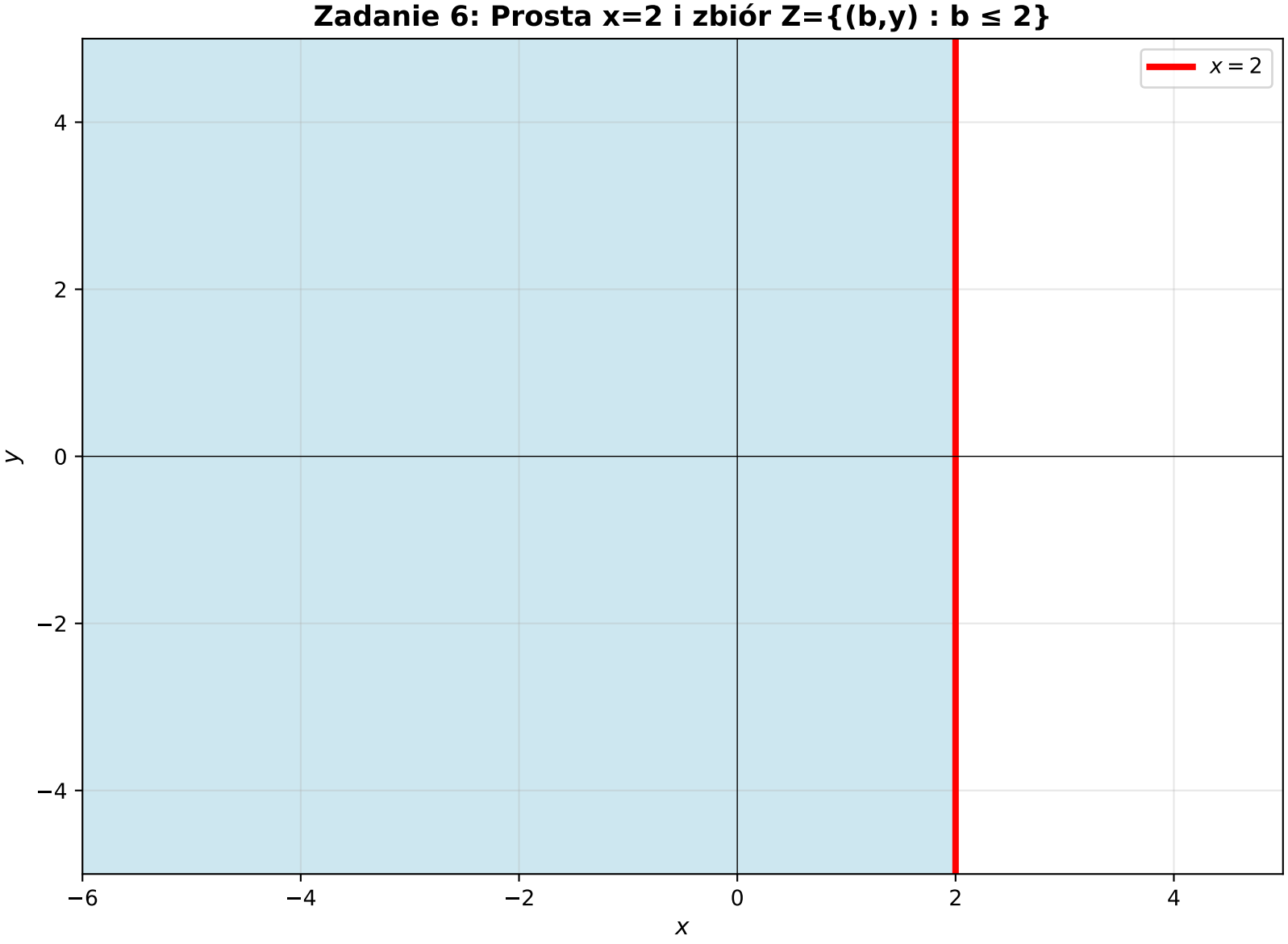

Zadanie 2: Trzy funkcje na jednym wykresie
$f_1(x) = x^2, x \in [-10, -1]$
$f_2(x) = x^3, x \in [-5, 3]$
$f_3(x) = x + 1, x \in [4, 10]$
100
50
0
−50
−100
y
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x

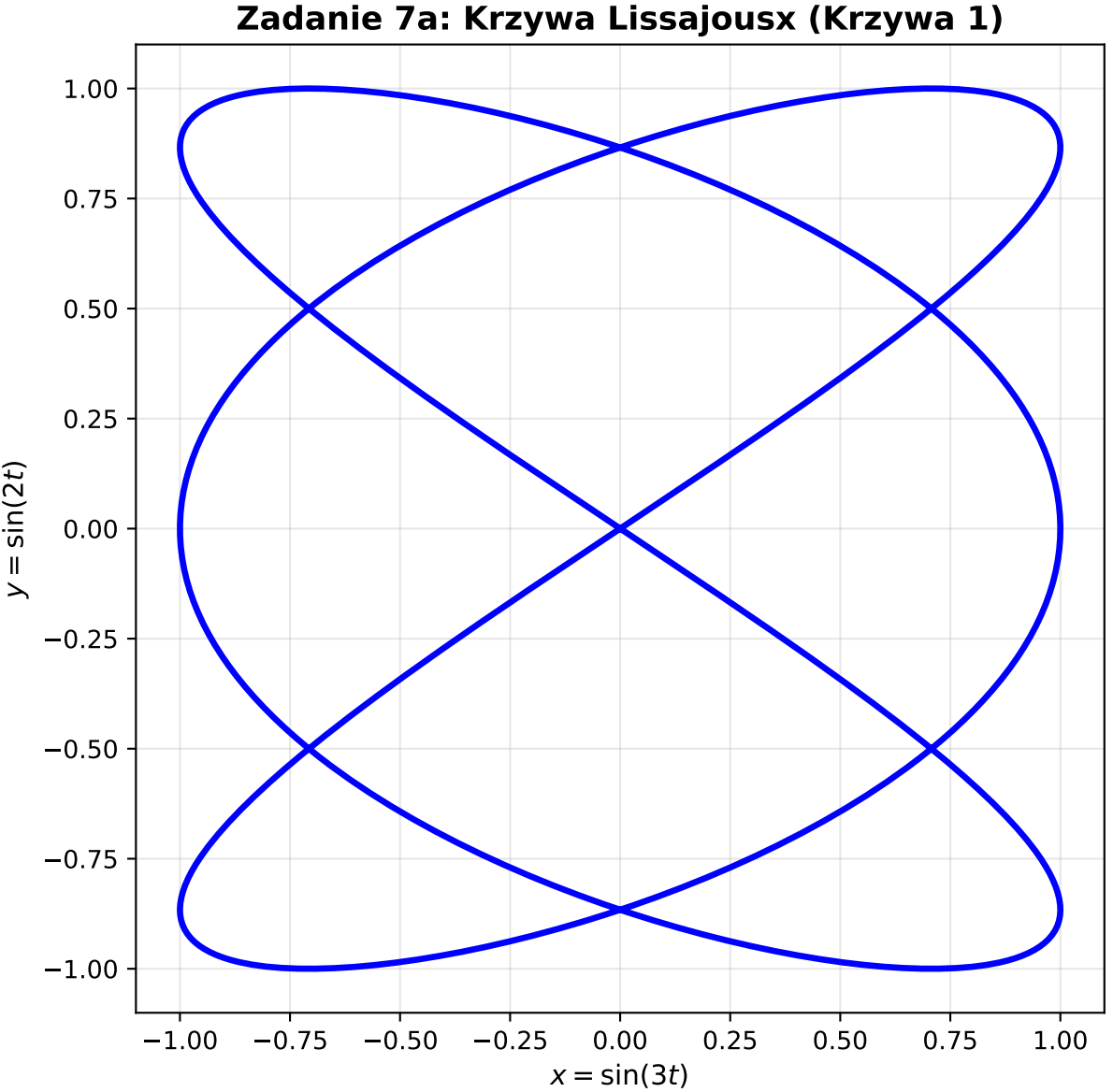

Zadanie 3: Funkcja klamerkowa i jej pochodna
f(x) dla x < 5: -3x + 4
f(x) dla x > 5: 1 + x^2
f'(x) dla x < 5: -3
f'(x) dla x > 5: 2x
y
x
100
80
60
40
20
0
0
2
4
6
8
10

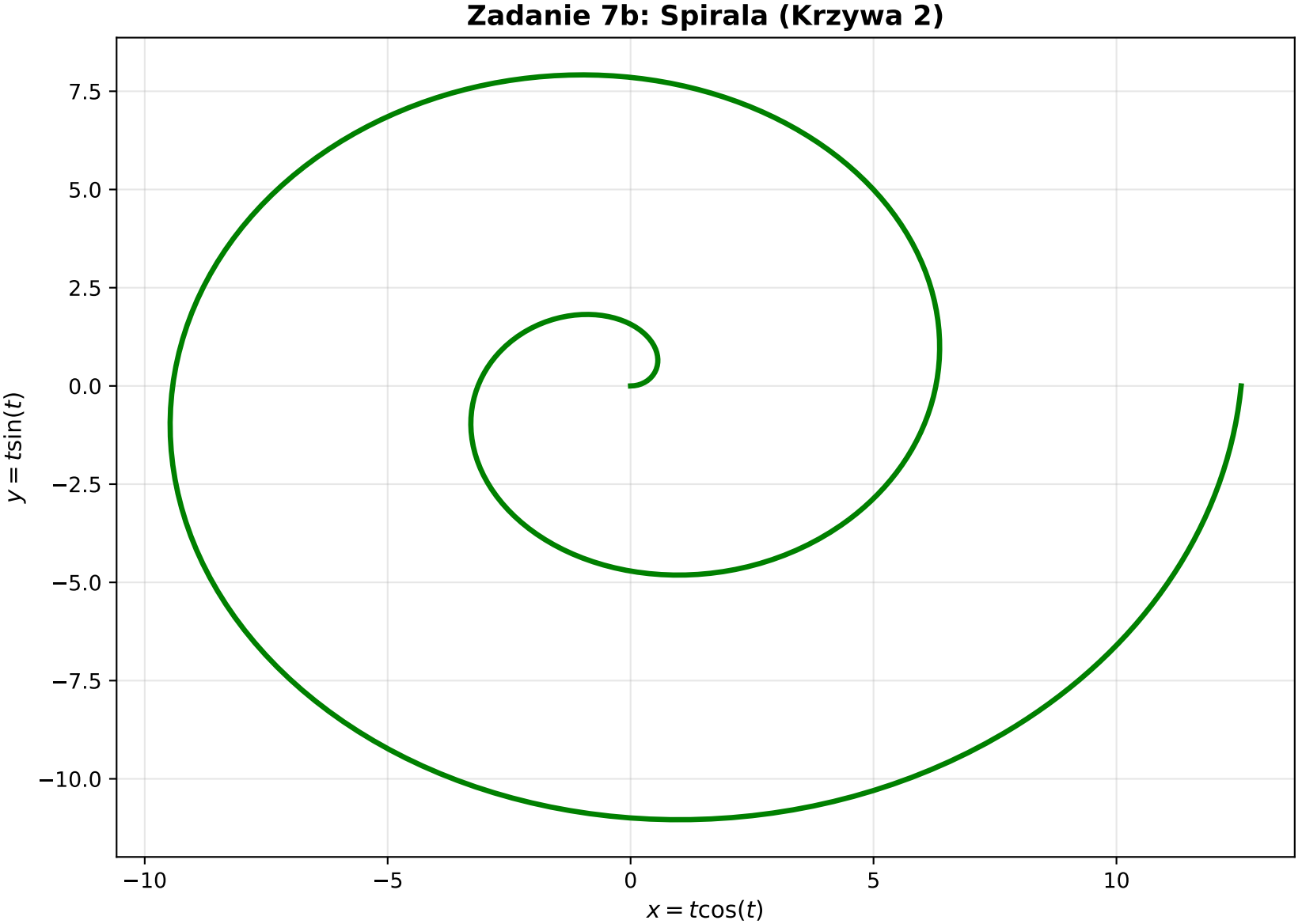

Zadanie 4: Trójkąt z trzema prostymi
$y = 2x$
$y = -x + 6$
$x = 1$
x
y
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
1
2
3
4
5
6
7

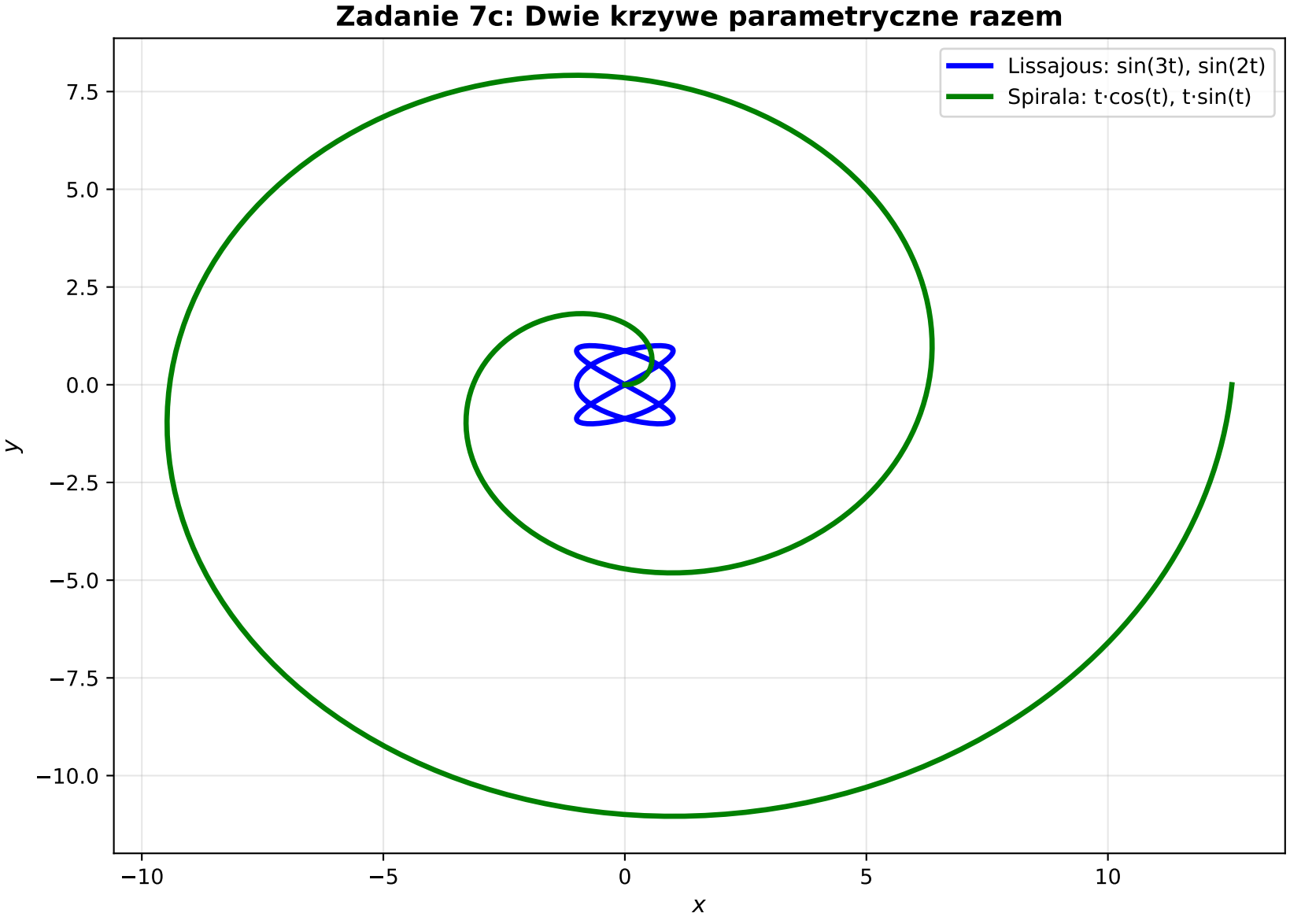
Zadanie 5: Krzywa uwikłana (elipsa) $\frac{x^2}{4} + \frac{y^2}{9} = 1$



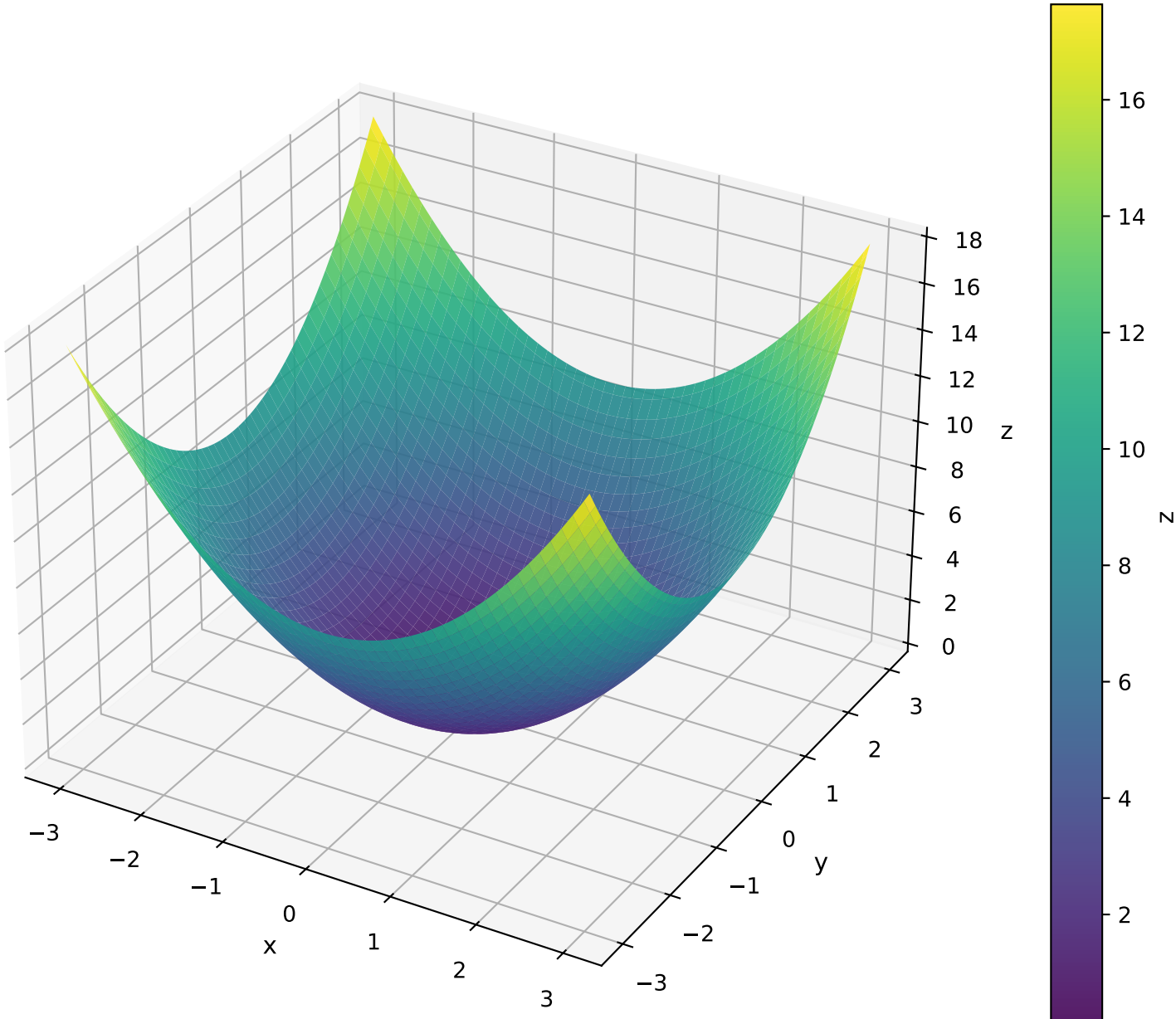

Zadanie 6: Prosta x=2 i zbiór Z={(b,y) : b ≤ 2}
x = 2
4
2
0
−2
−4
y
−6
−4
−2
0
2
4
x


Zadanie 7a: Krzywa Lissajousx (Krzywa 1)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
$y = \sin(2t)$
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
$x = \sin(3t)$

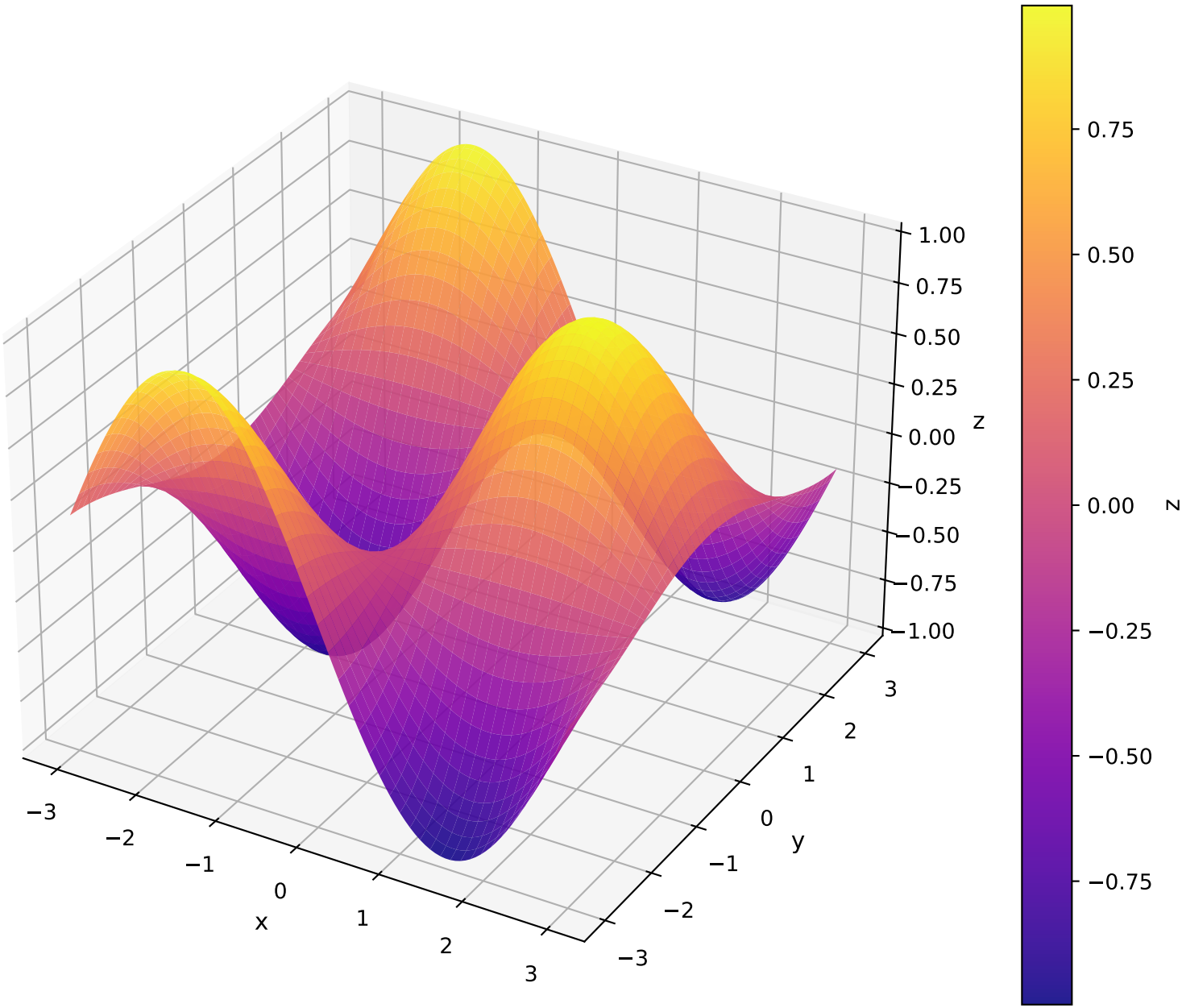

Zadanie 7b: Spirala (Krzywa 2)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
$y = t\sin(t)$
−10
−5
0
5
10
$x = t\cos(t)$


Zadanie 7c: Dwie krzywe parametryczne razem
Lissajous: sin(3t), sin(2t)
Spirala: t·cos(t), t·sin(t)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y
−10
−5
0
5
10
x

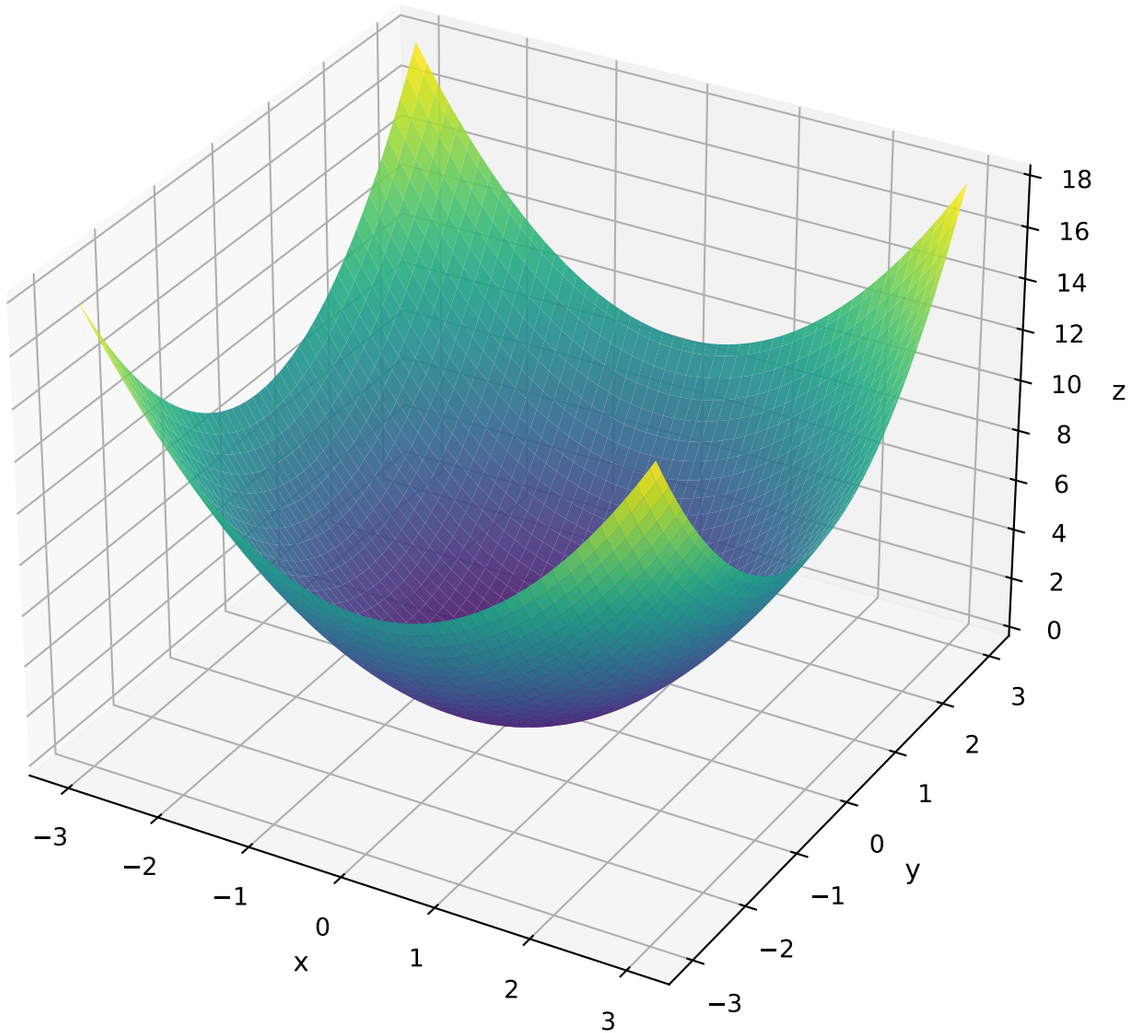
Zadanie 8a: Paraboloida $z = x^2 + y^2$



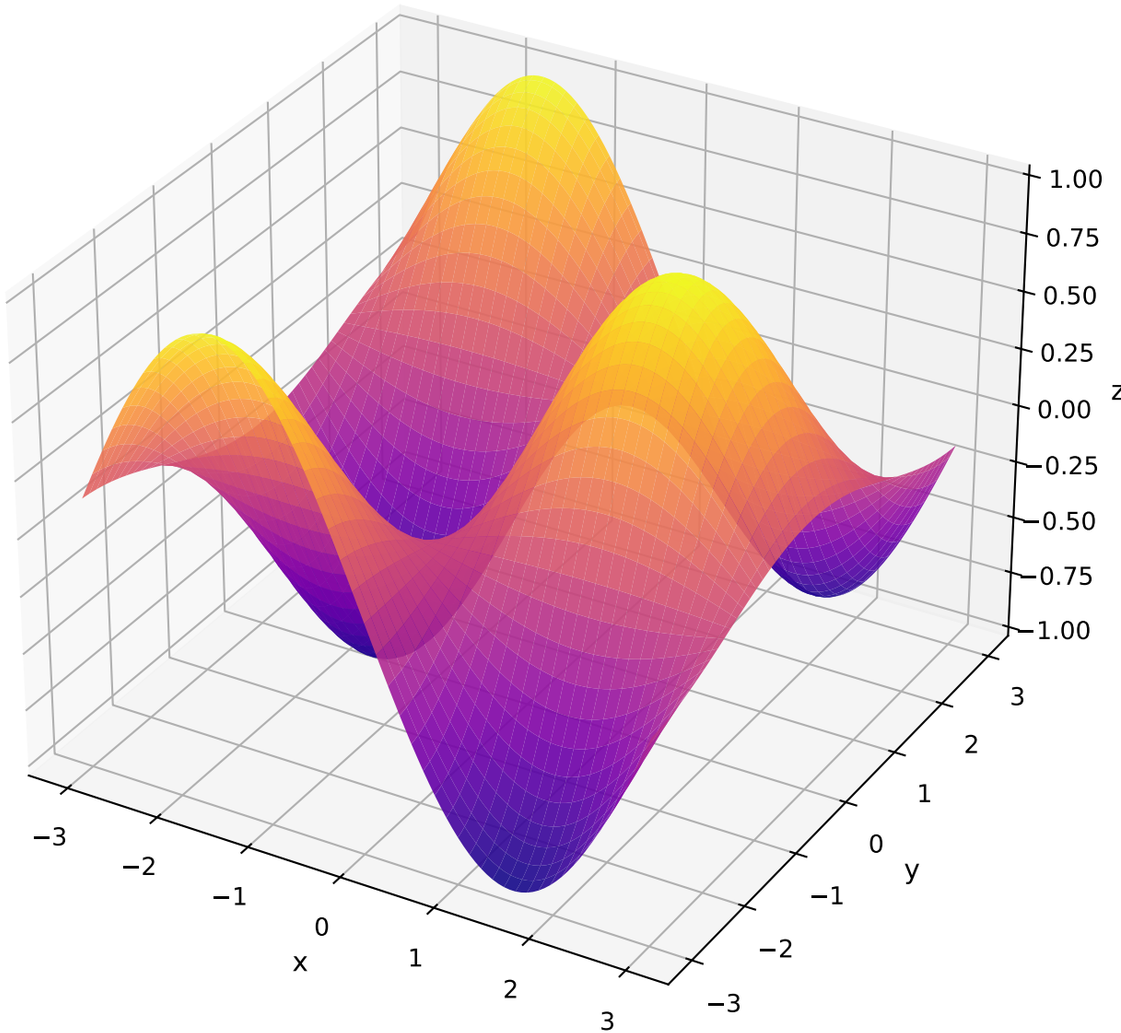
Zadanie 8b: Funkcja $z = \sin(x)\cos(y)$

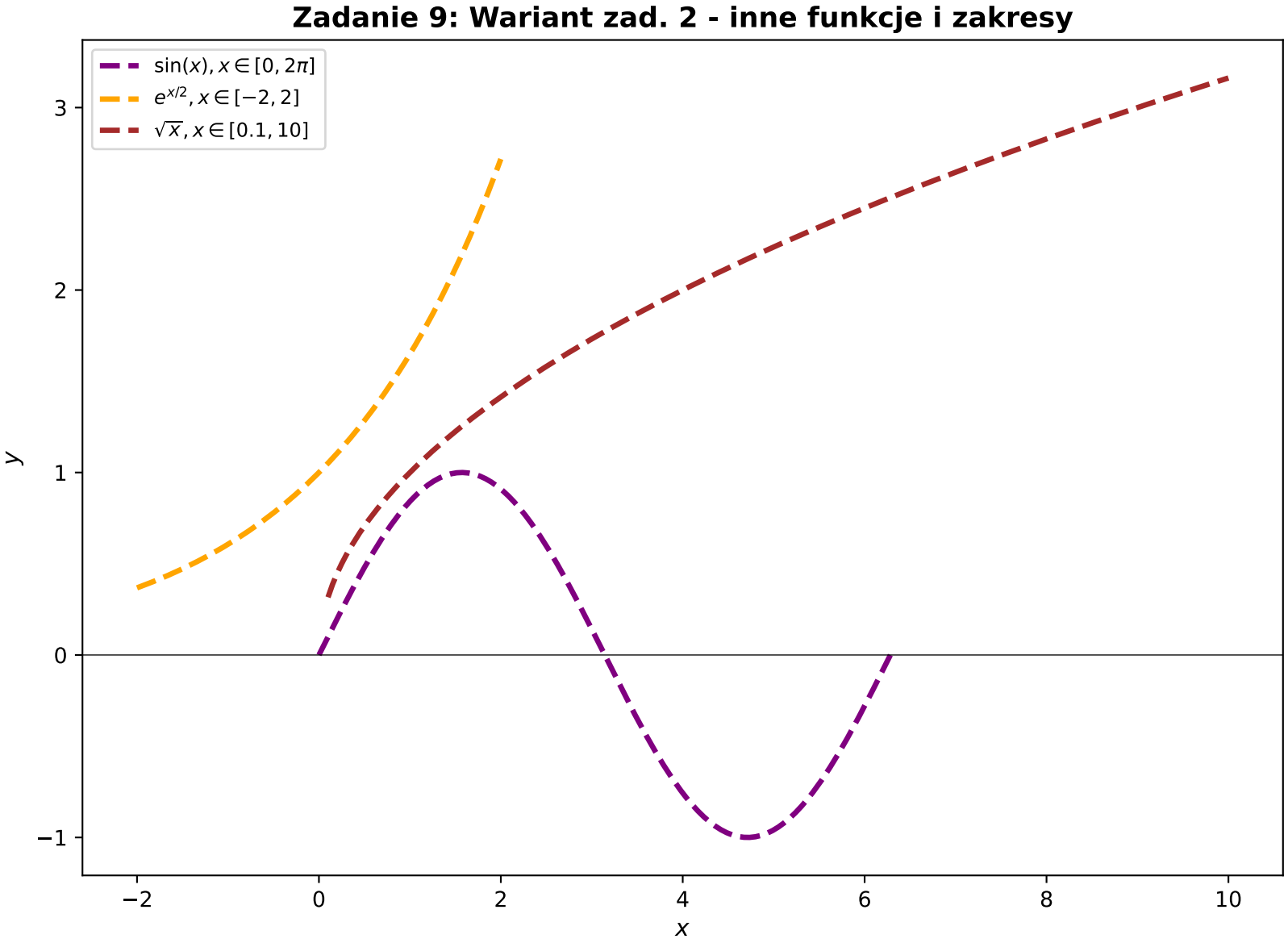


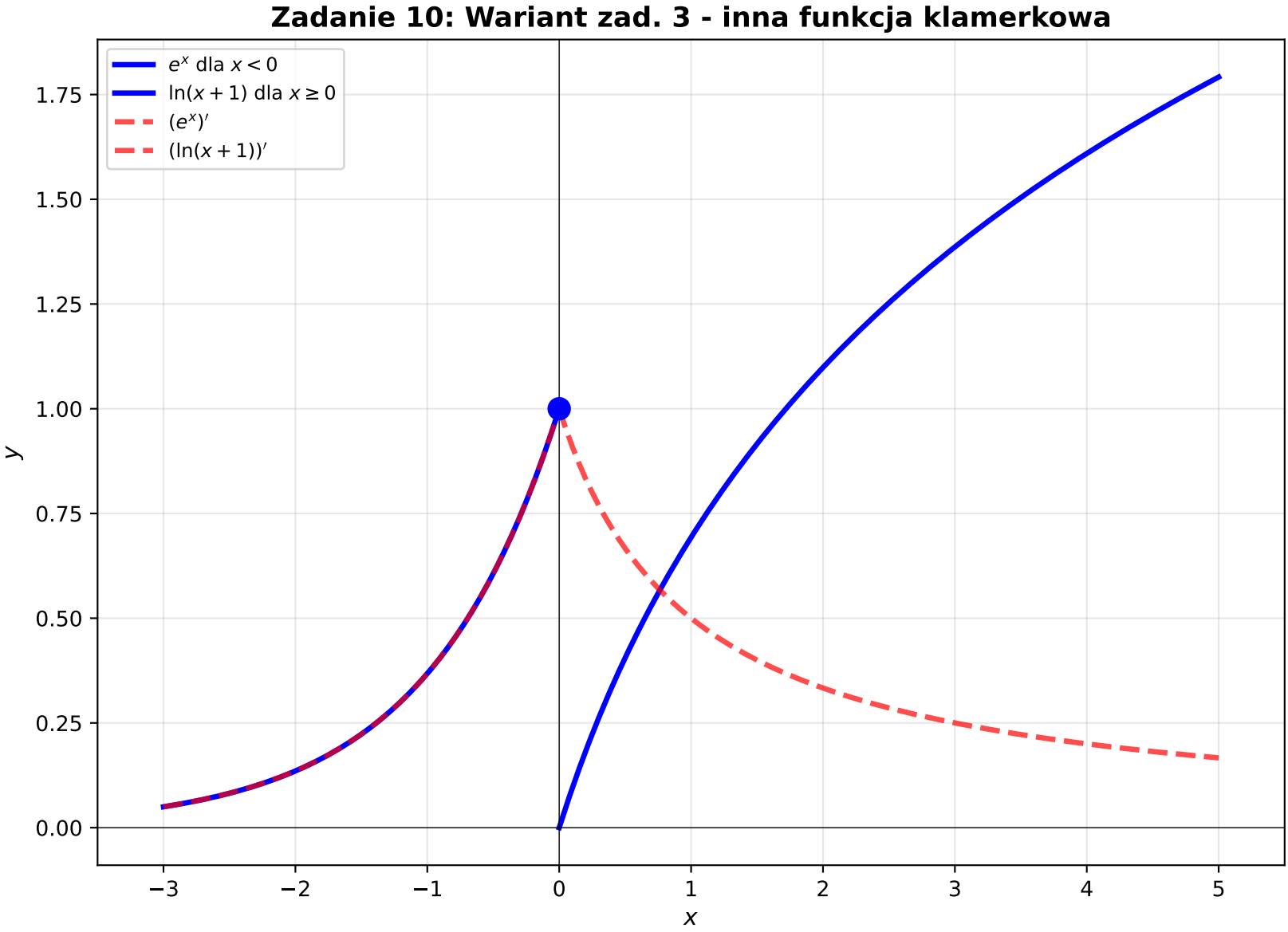
Paraboloida: $z = x^2 + y^2$

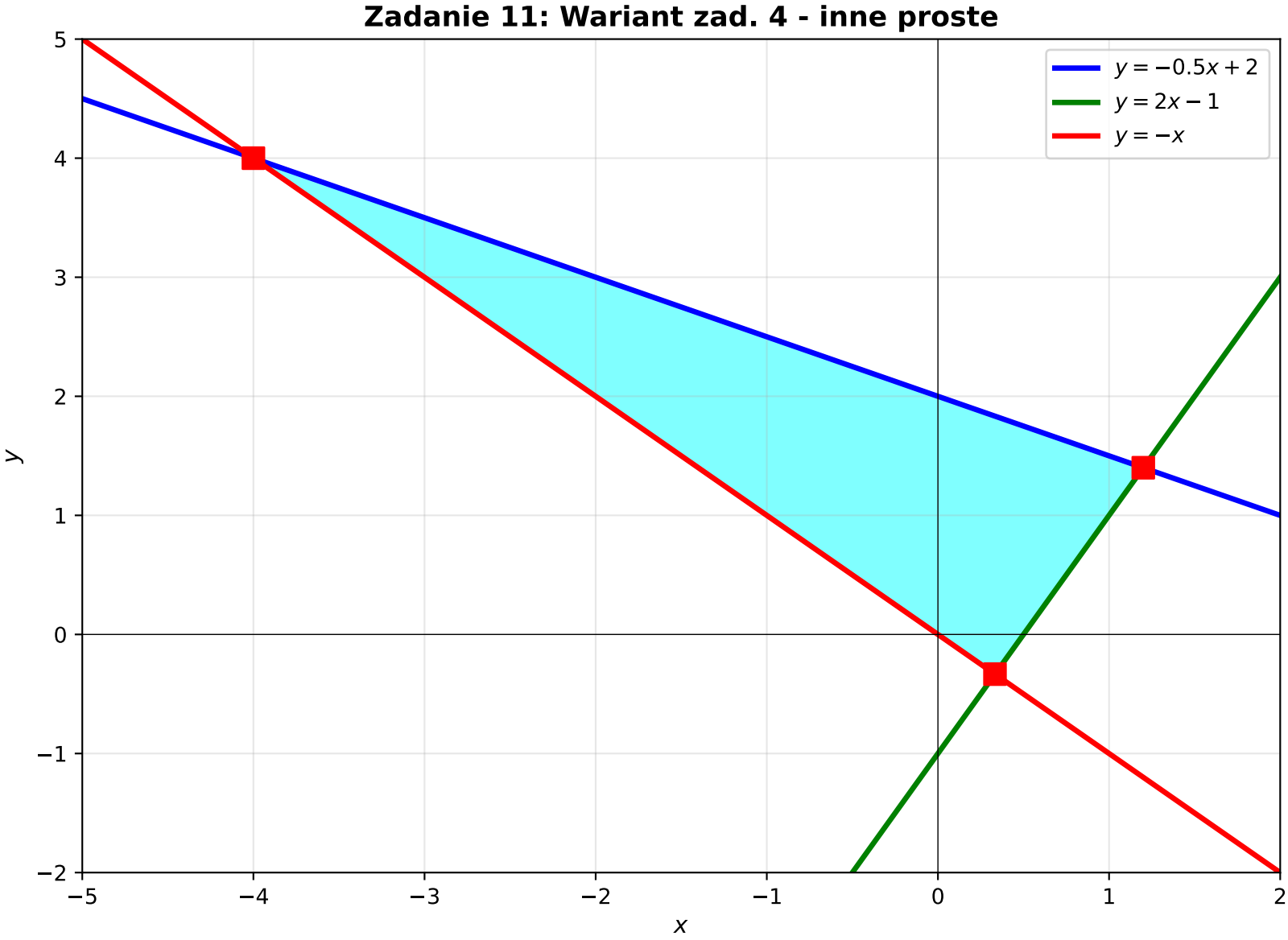


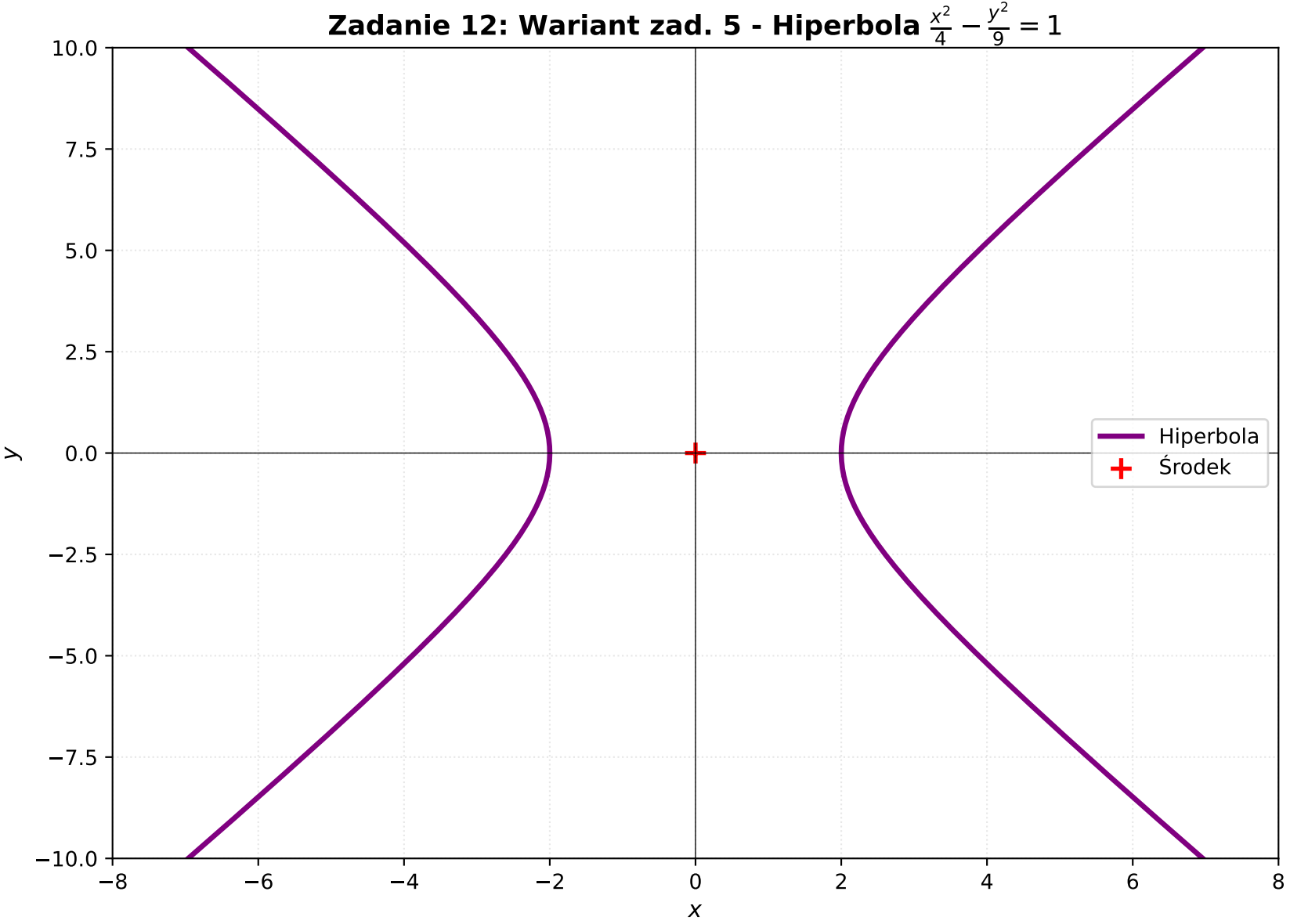
Funkcja: $z = \sin(x)\cos(y)$

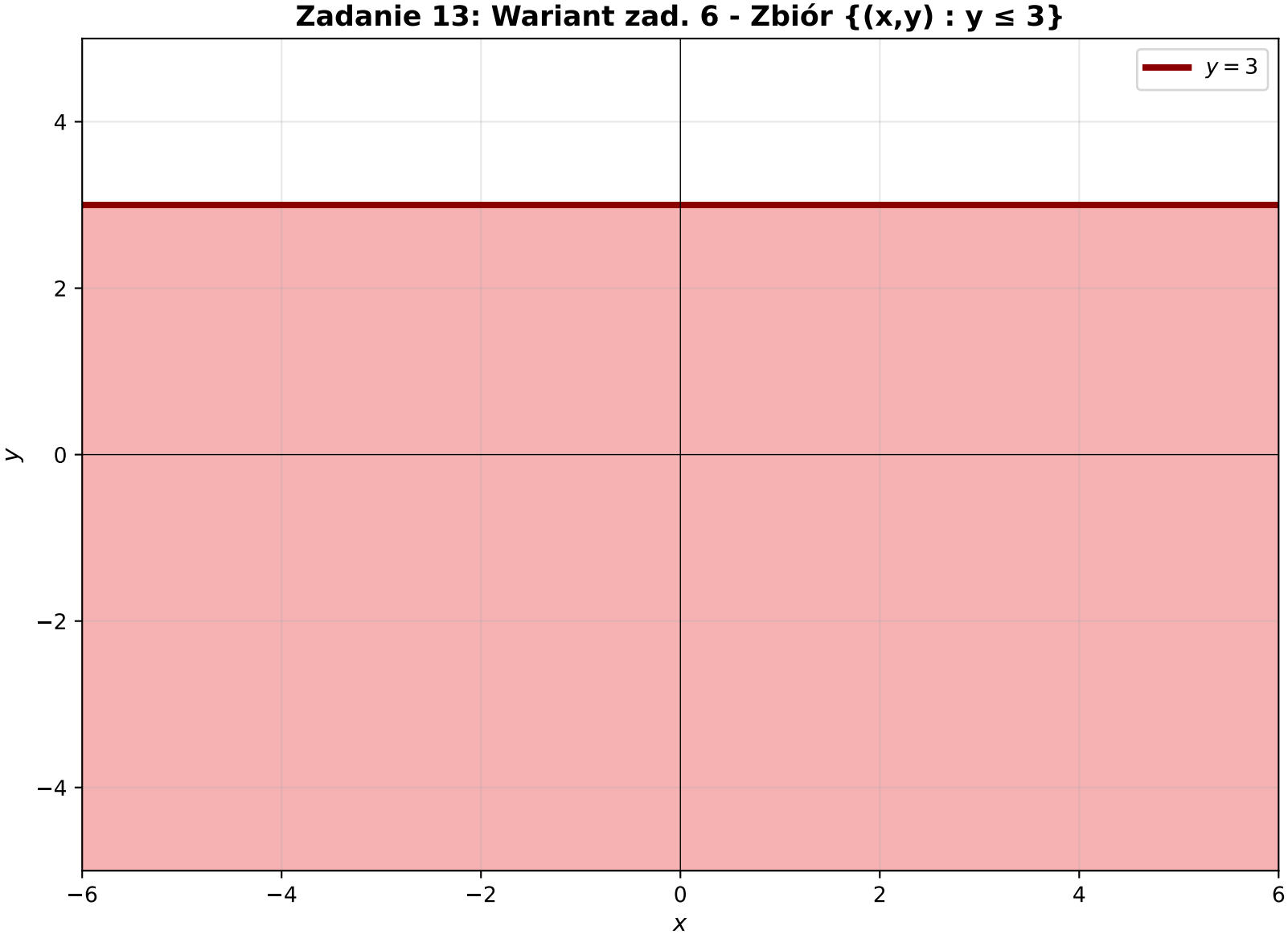


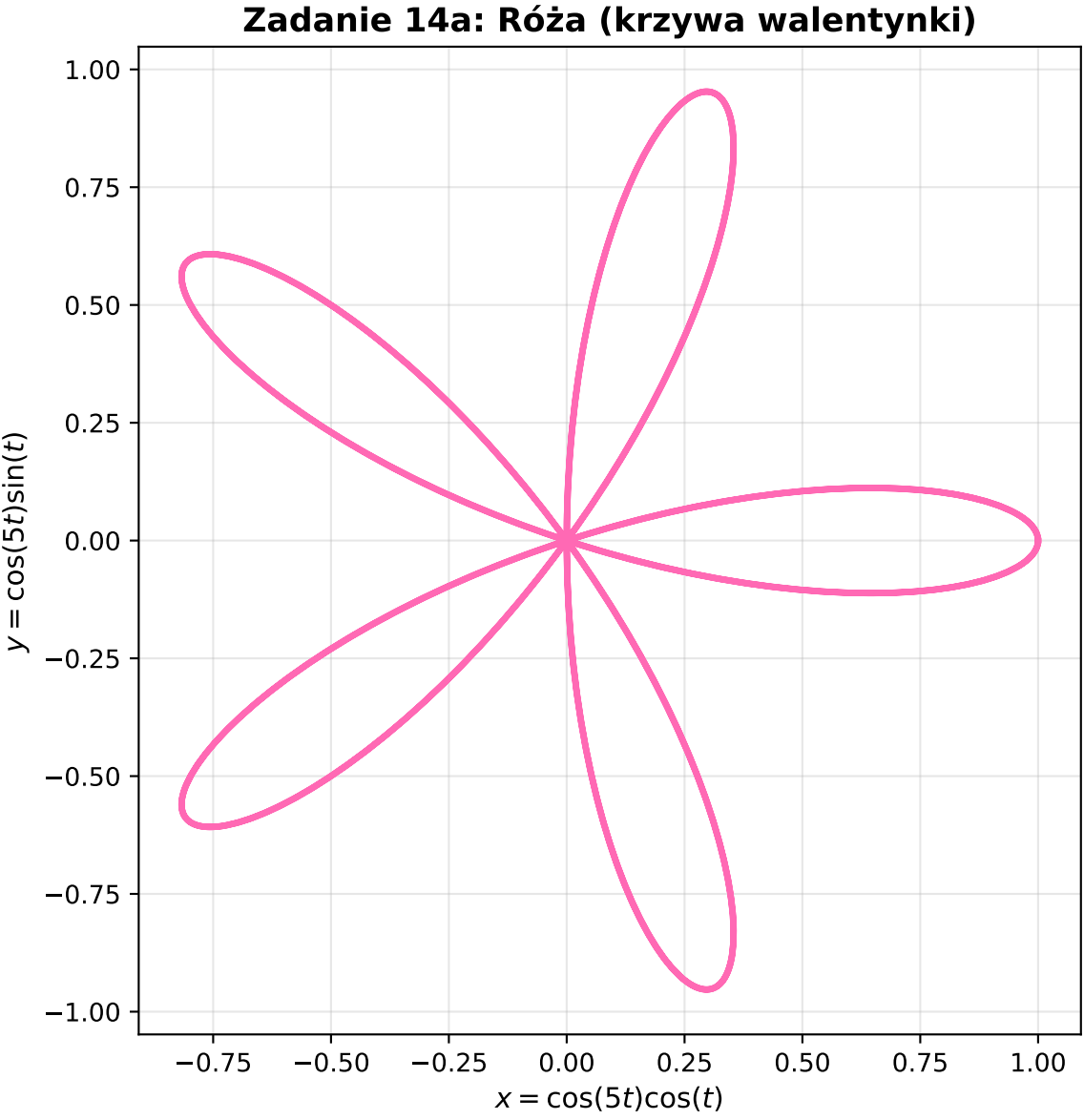

Zadanie 9: Wariant zad. 2 - inne funkcje i zakresy
$\sin(x), x \in [0, 2\pi]$
$e^{x/2}, x \in [-2, 2]$
$\sqrt{x}, x \in [0.1, 10]$
3
2
1
0
−1
y
−2
0
2
4
6
8
10
x

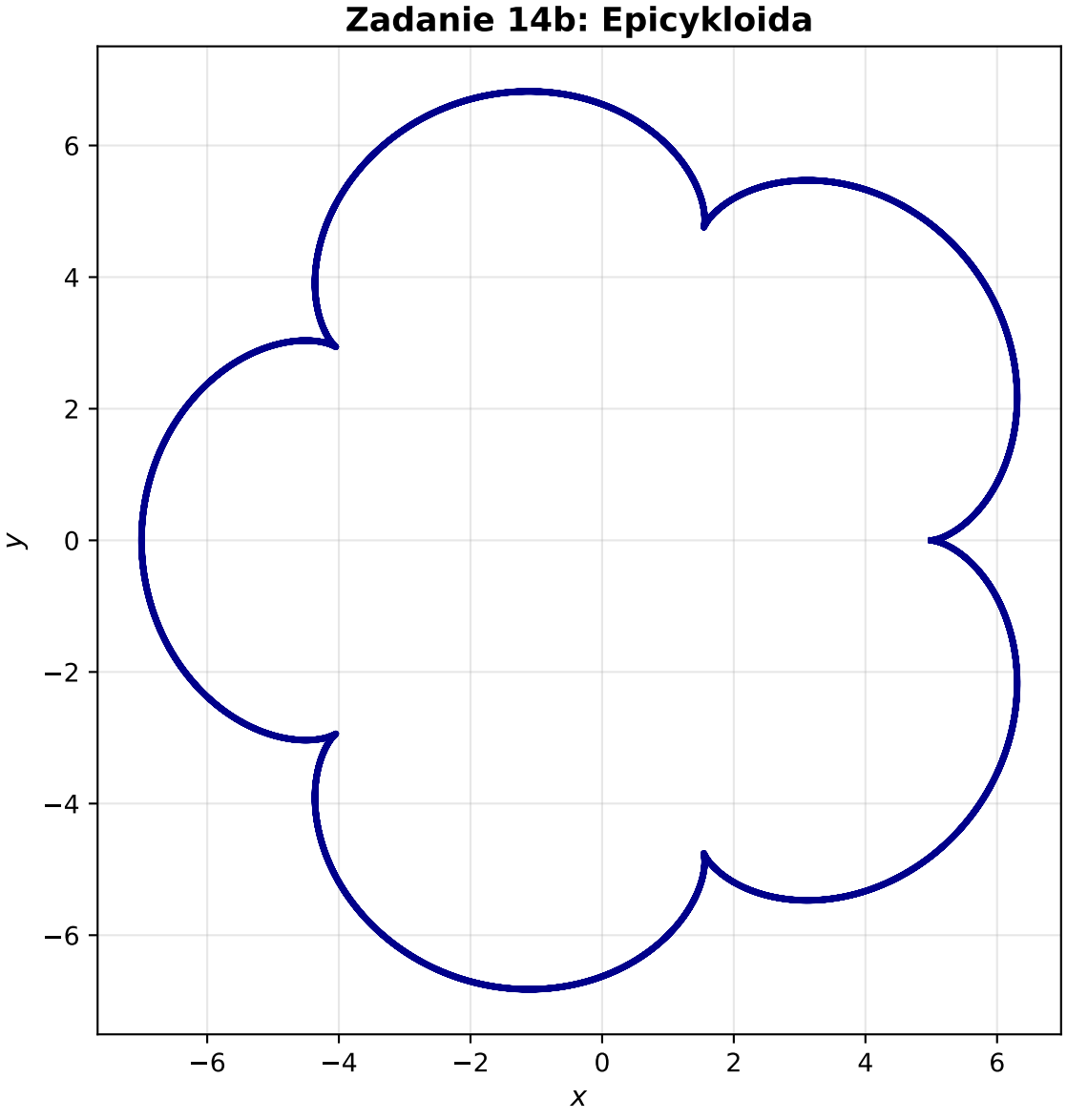

Zadanie 10: Wariant zad. 3 - inna funkcja klamerkowa
e^x dla $x < 0$
$\ln(x+1)$ dla $x \geq 0$
$(e^x)'$
$(\ln(x+1))'$
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
y
−3
−2
−1
0
1
2
3
4
5
x

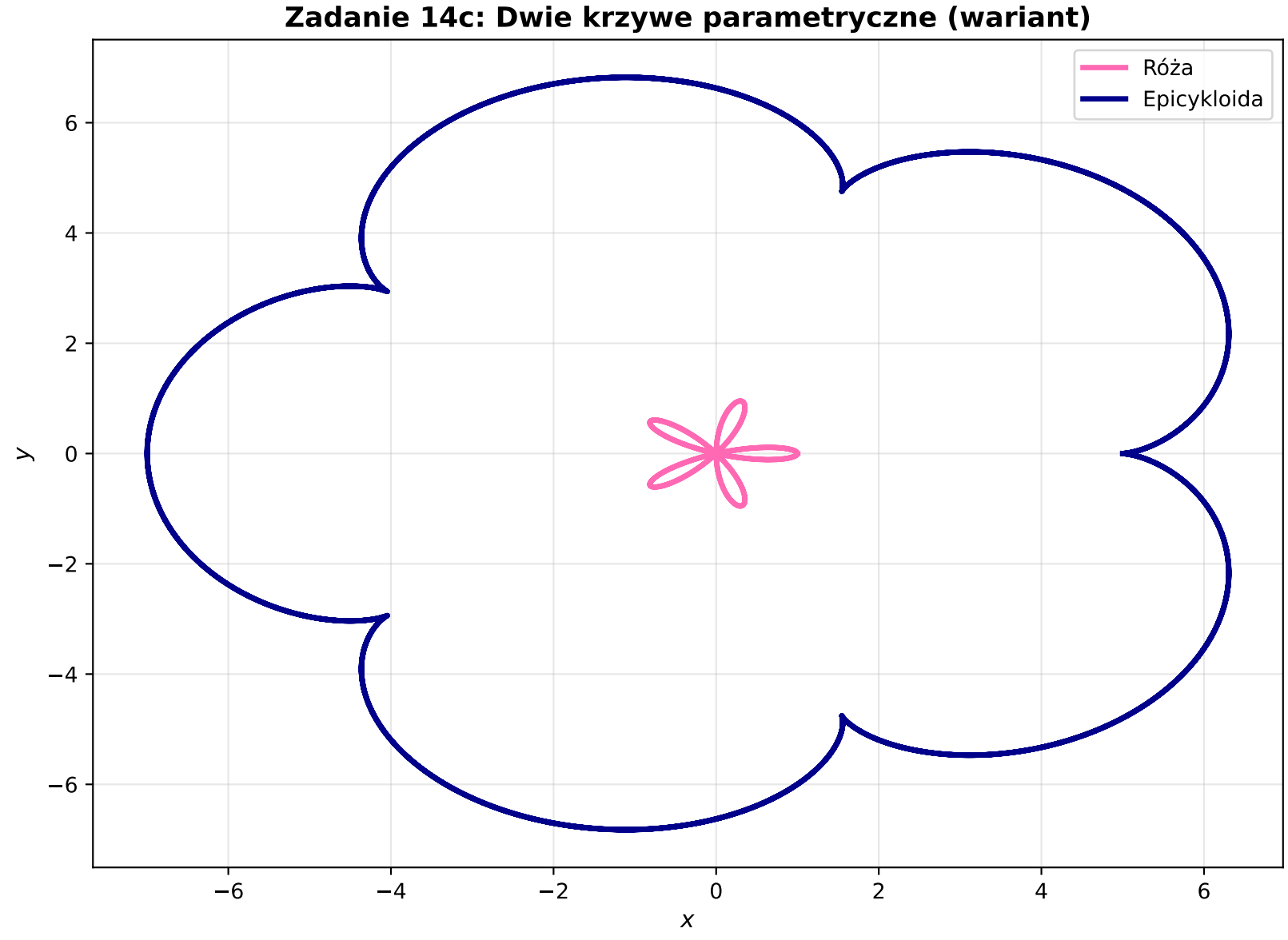

Zadanie 11: Wariant zad. 4 - inne proste
$y = -0.5x + 2$
$y = 2x - 1$
$y = -x$
x
y
5
4
3
2
1
0
−1
−2
−5
−4
−3
−2
−1
0
1
2


Zadanie 12: Wariant zad. 5 - Hiperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y
−8
−6
−4
−2
0
2
4
6
8
x
Hiperbola
Środek

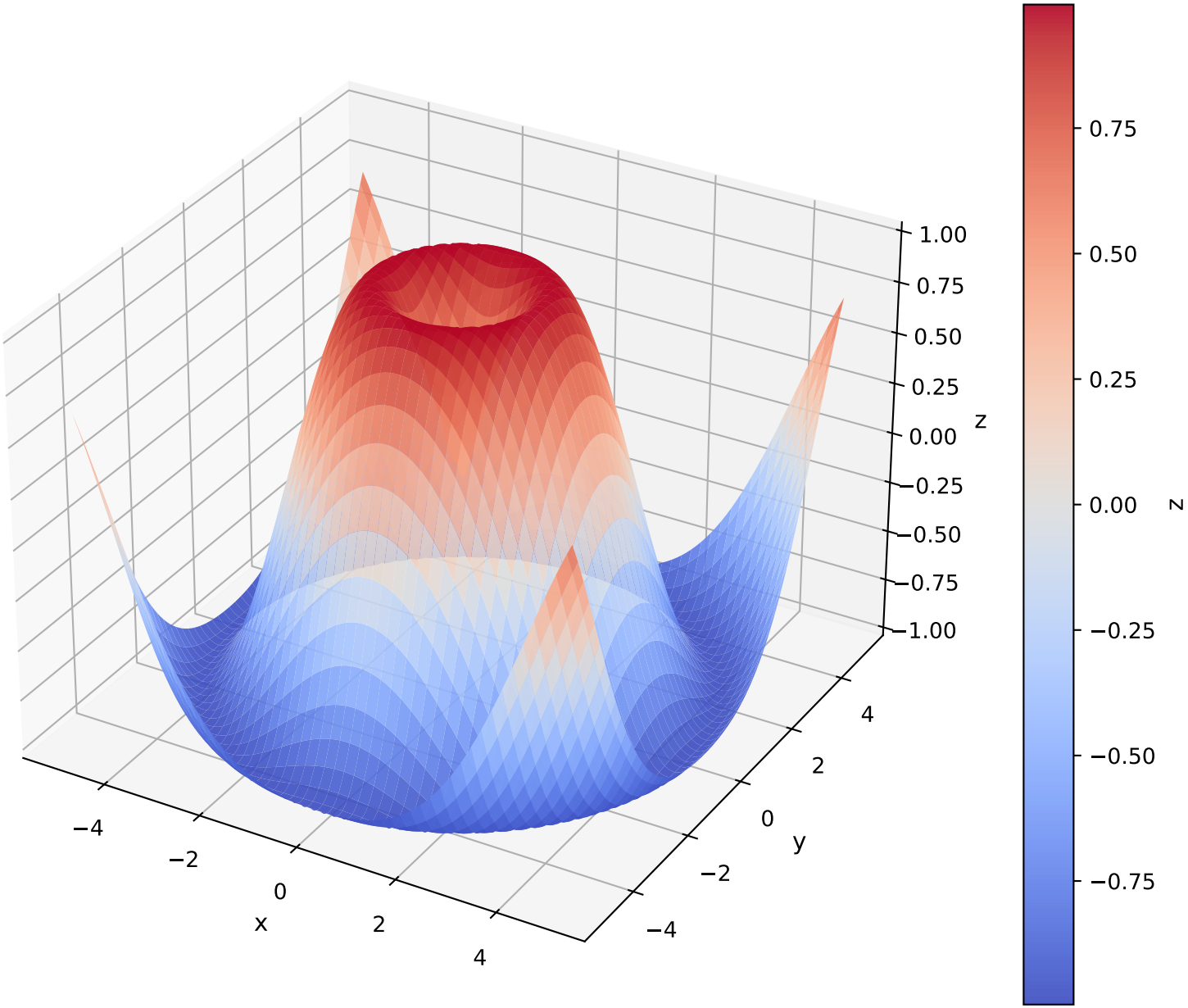

Zadanie 13: Wariant zad. 6 - Zbiór {(x,y) : y ≤ 3}
y = 3
4
2
0
−2
−4
y
−6
−4
−2
0
2
4
6
x


Zadanie 14a: Róża (krzywa walentynki)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
$y = \cos(5t)\sin(t)$
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
$x = \cos(5t)\cos(t)$

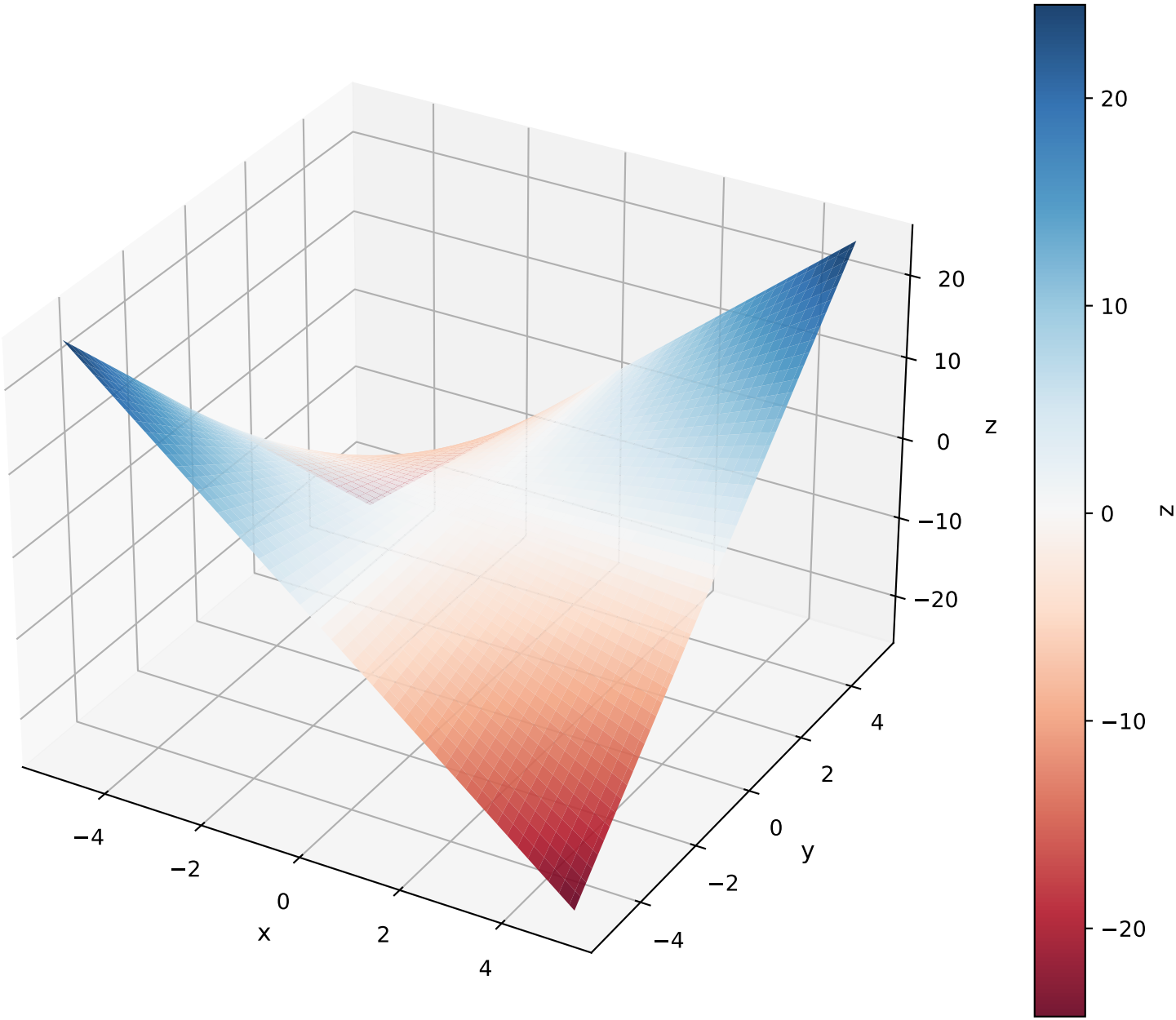

Zadanie 14b: Epicykloida
6
4
2
0
−2
−4
−6
y
−6
−4
−2
0
2
4
6
x


Zadanie 14c: Dwie krzywe parametryczne (wariant)
Róża
Epicykloida
6
4
2
0
−2
−4
−6
y
−6
−4
−2
0
2
4
6
x

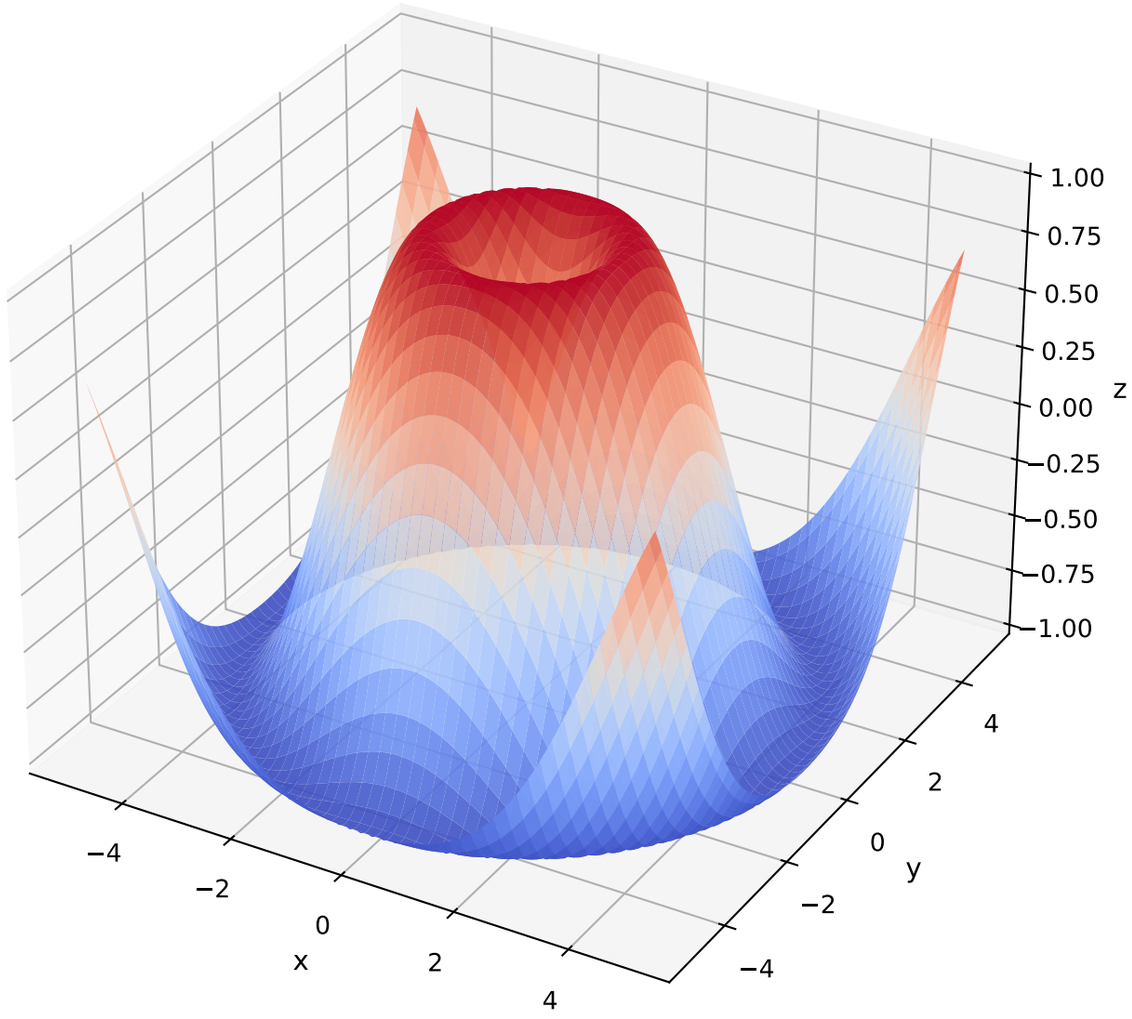
Zadanie 15a: Fale $z = \sin(\sqrt{x^2 + y^2})$



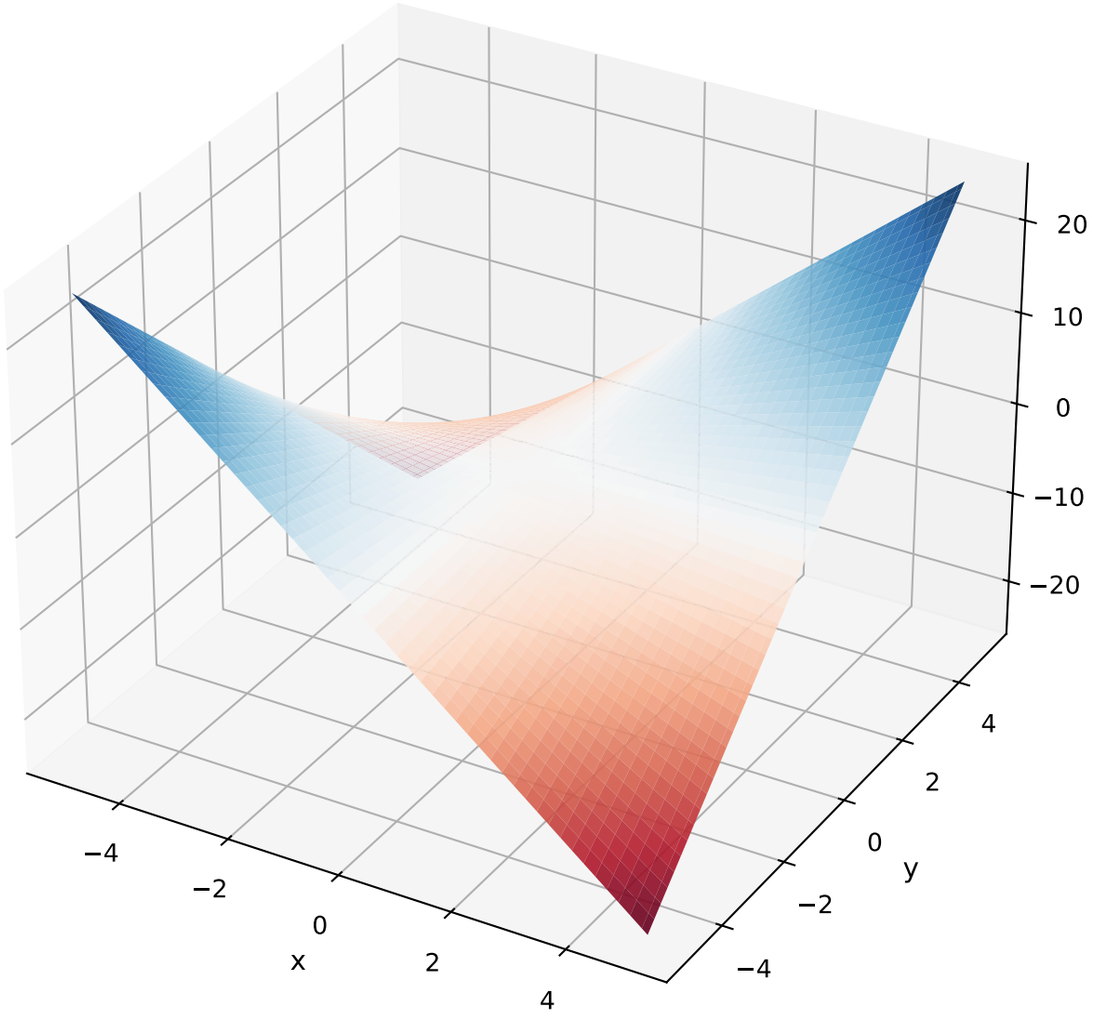
Zadanie 15b: Siodło $z = xy$

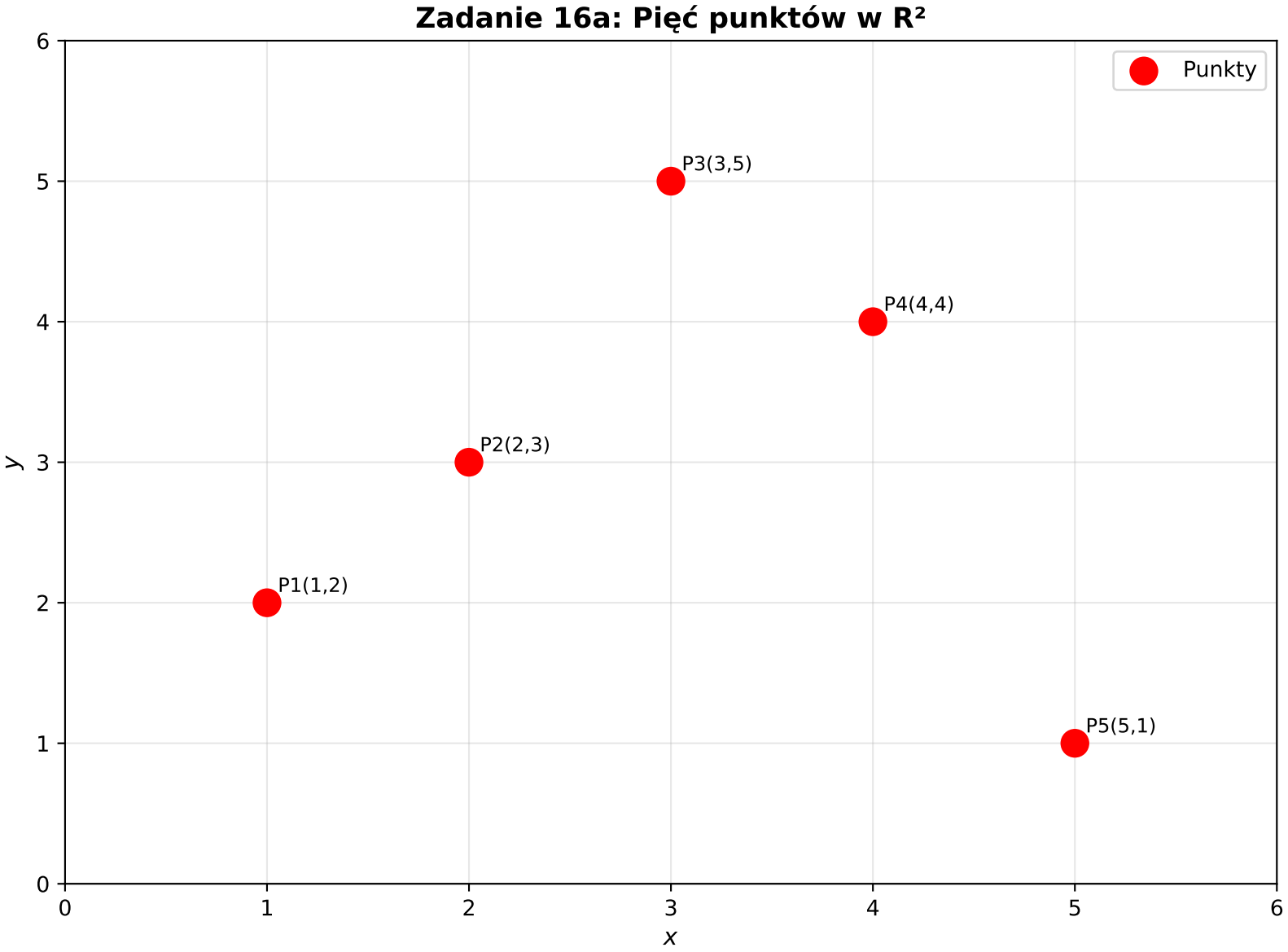


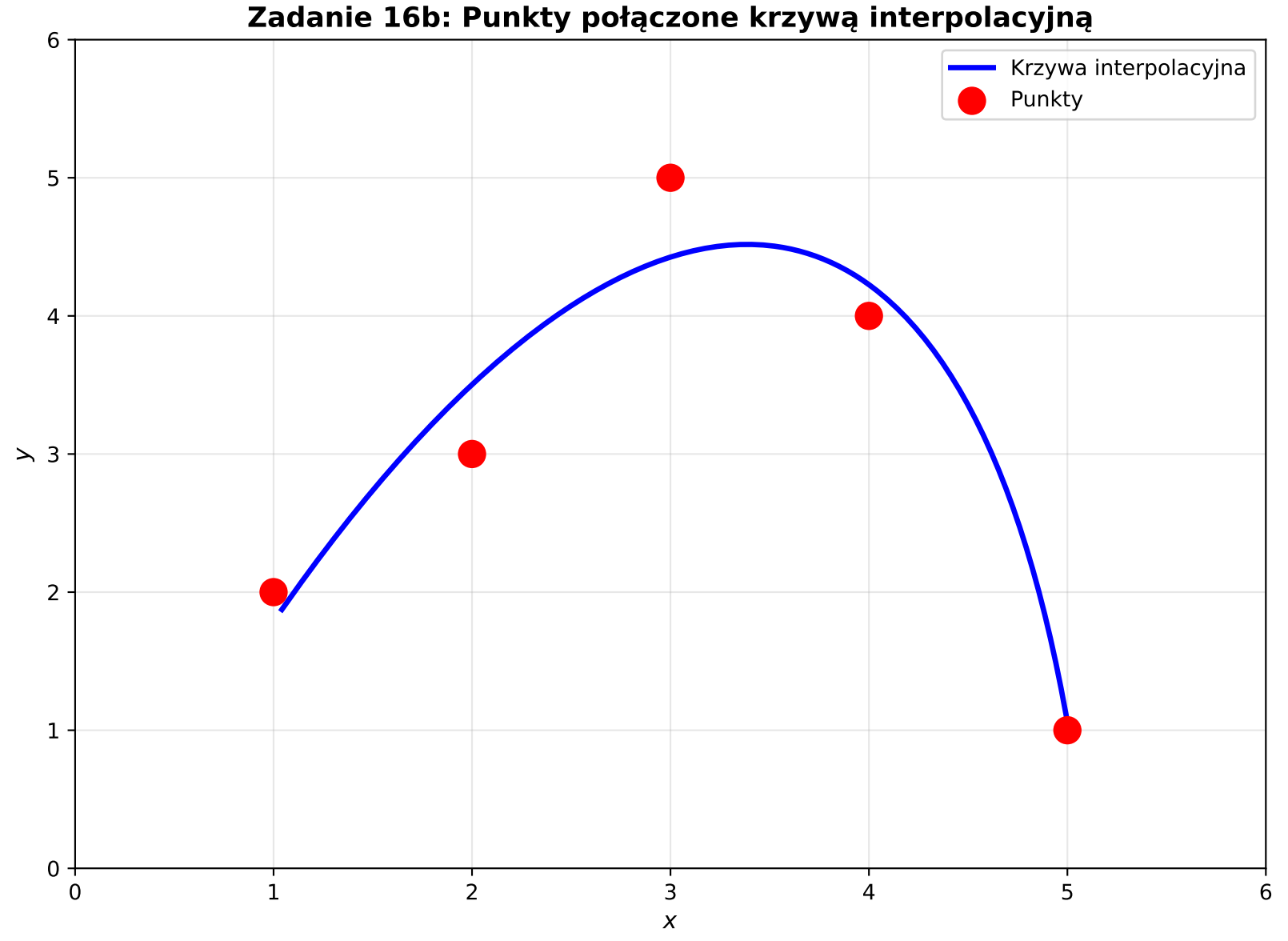
Fale: $z = \sin(\sqrt{x^2 + y^2})$



Siodło: $z = xy$

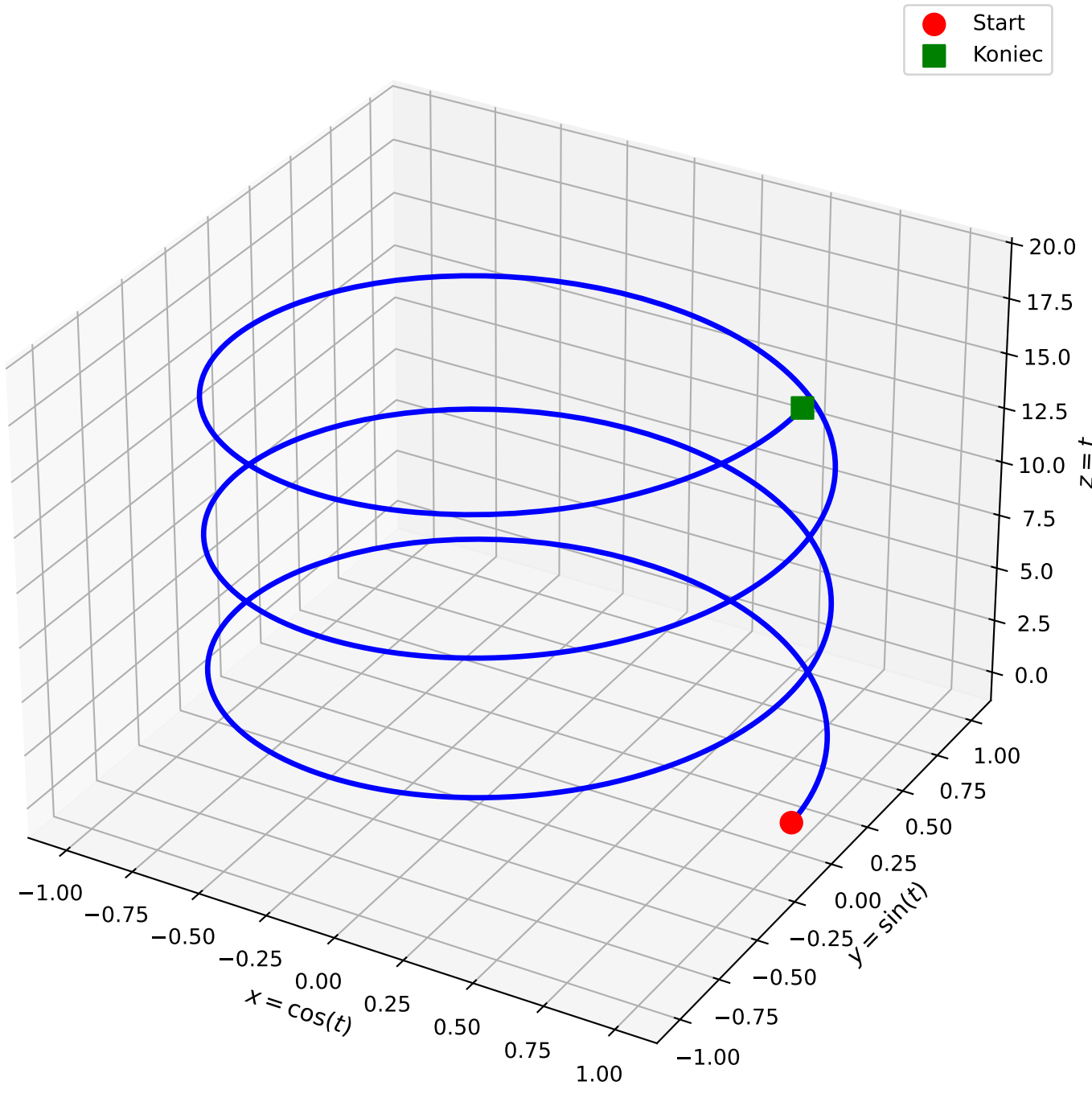



Zadanie 16a: Pięć punktów w R²
Punkty
P1(1,2)
P2(2,3)
P3(3,5)
P4(4,4)
P5(5,1)
x
y
0
1
2
3
4
5
6


Zadanie 16b: Punkty połączone krzywą interpolacyjną
Krzywa interpolacyjna
Punkty
y
x
0
1
2
3
4
5
6

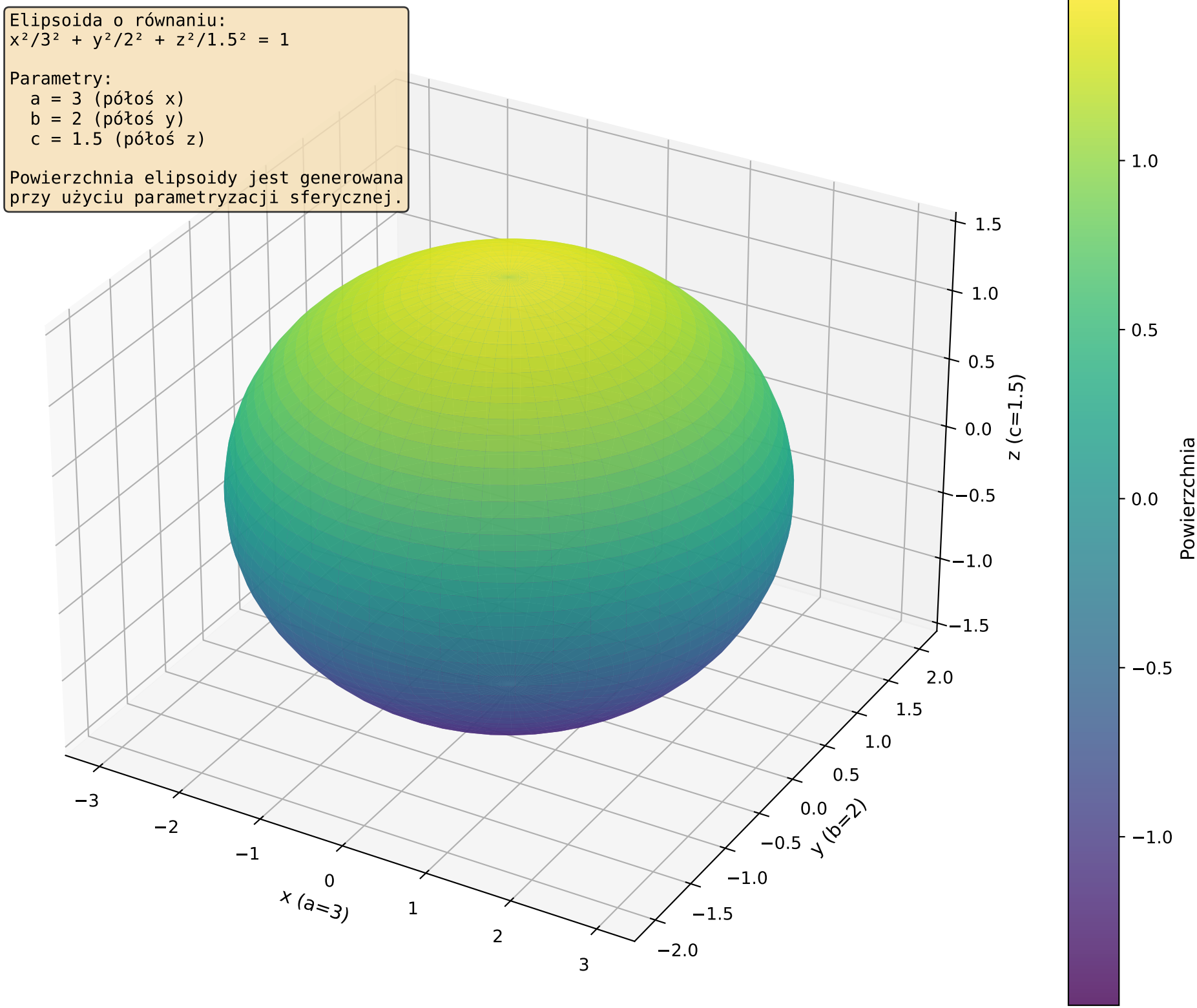
Zadanie 17: Krzywa 3D - Helisa

$x = \cos(t), y = \sin(t), z = t$

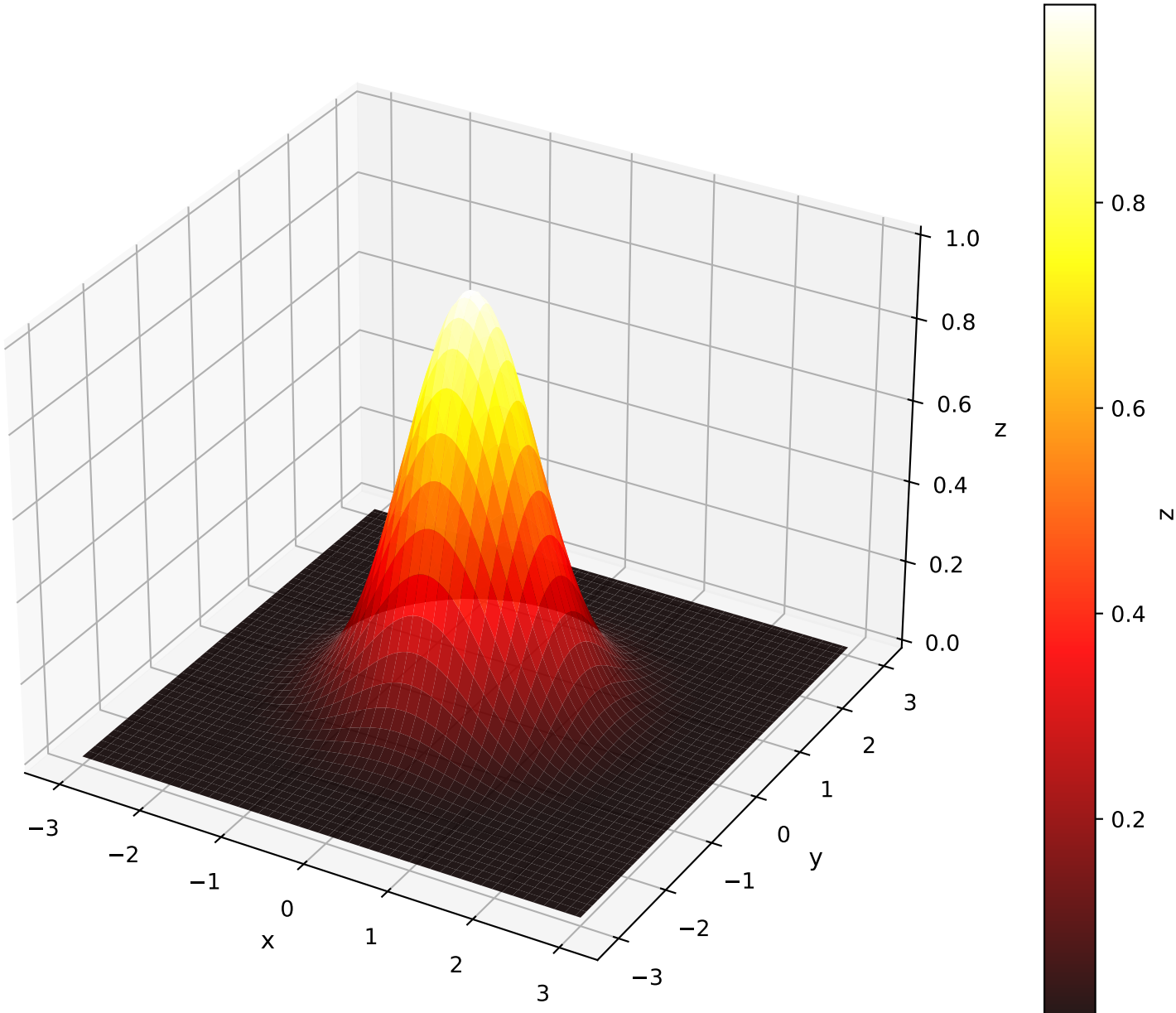


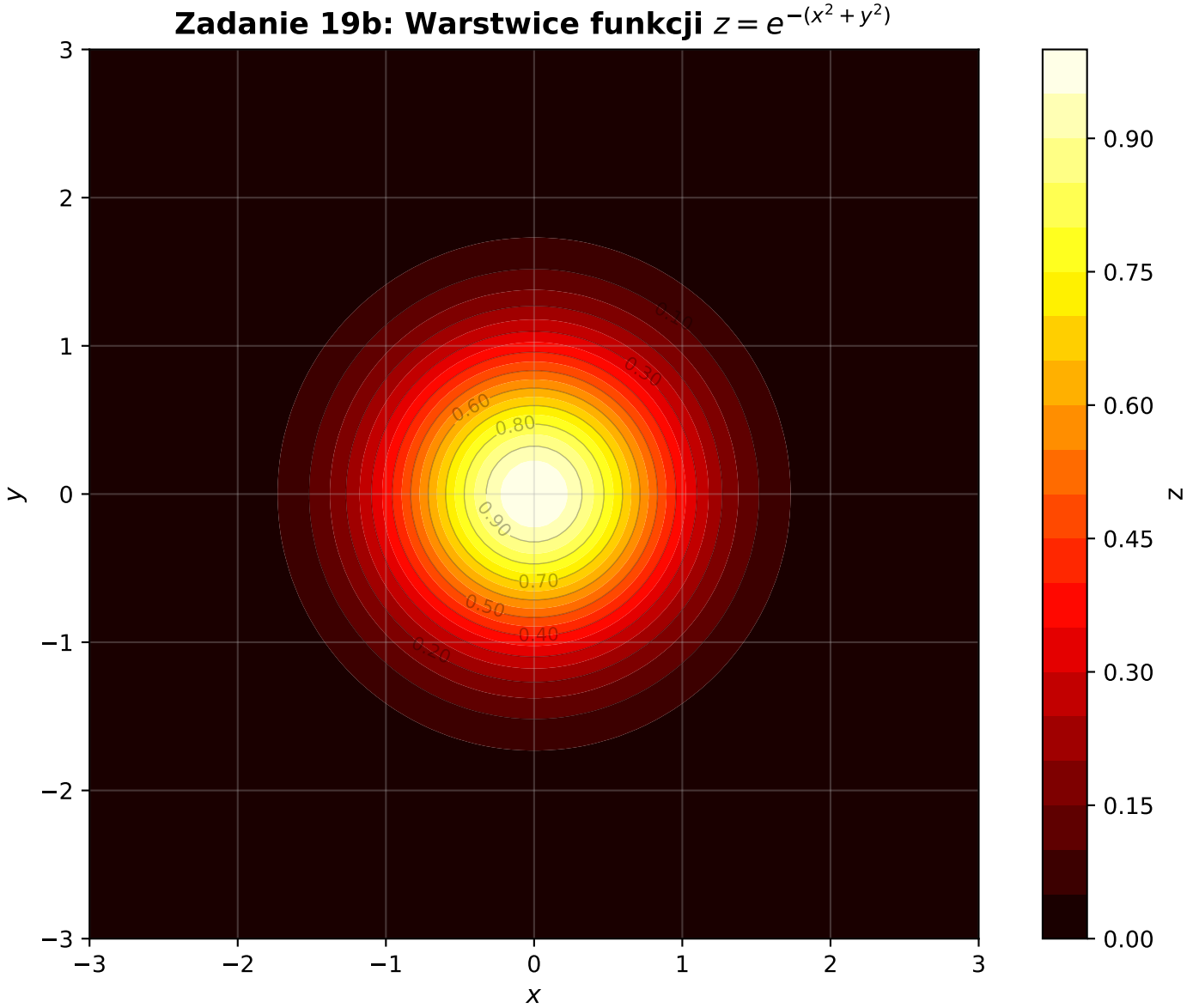
Zadanie 18: Elipsoida

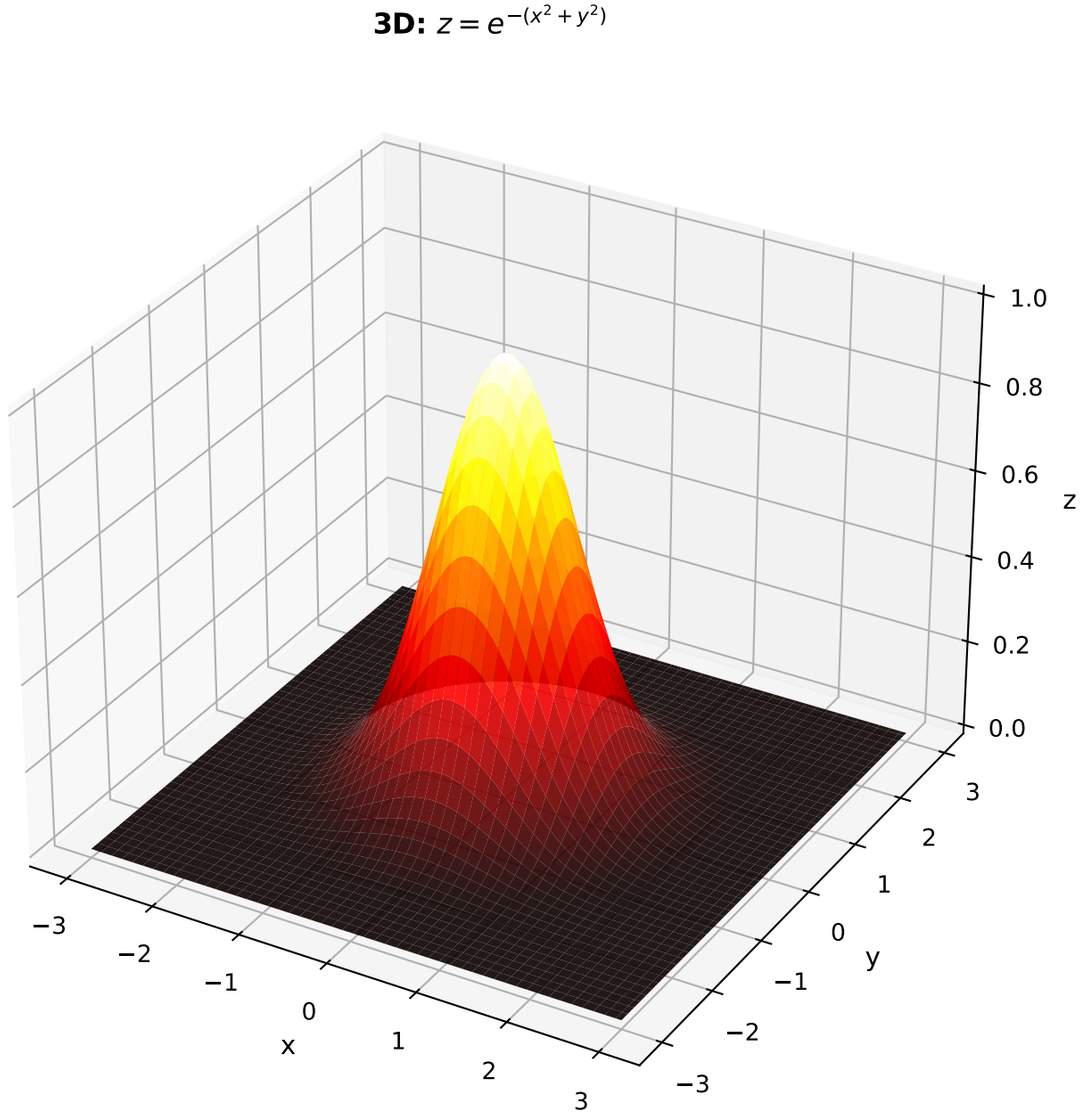
$$\frac{x^2}{3^2} + \frac{y^2}{2^2} + \frac{z^2}{1.5^2} = 1$$

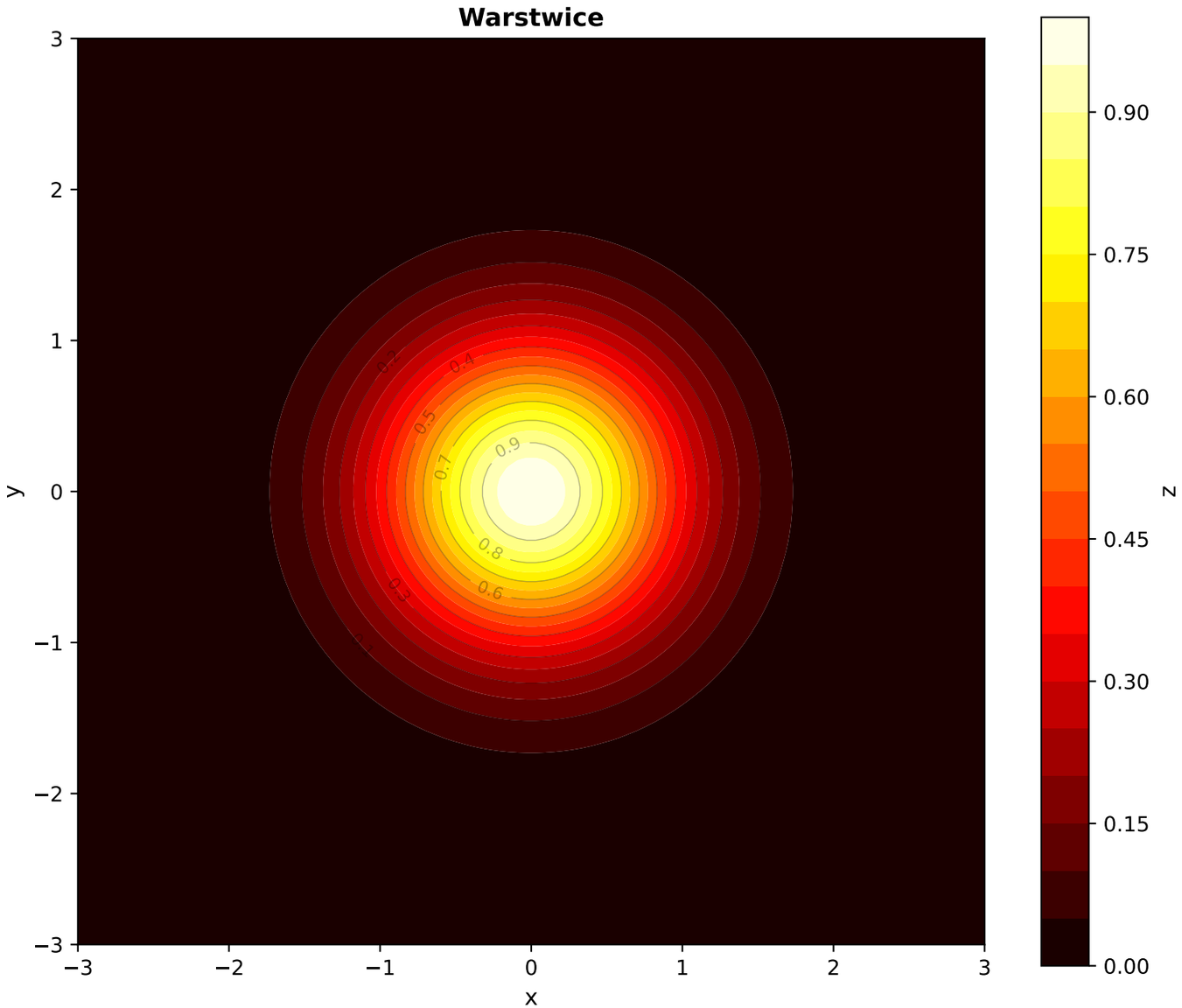


Zadanie 19a: Funkcja 2-zmiennych $z = e^{-(x^2+y^2)}$

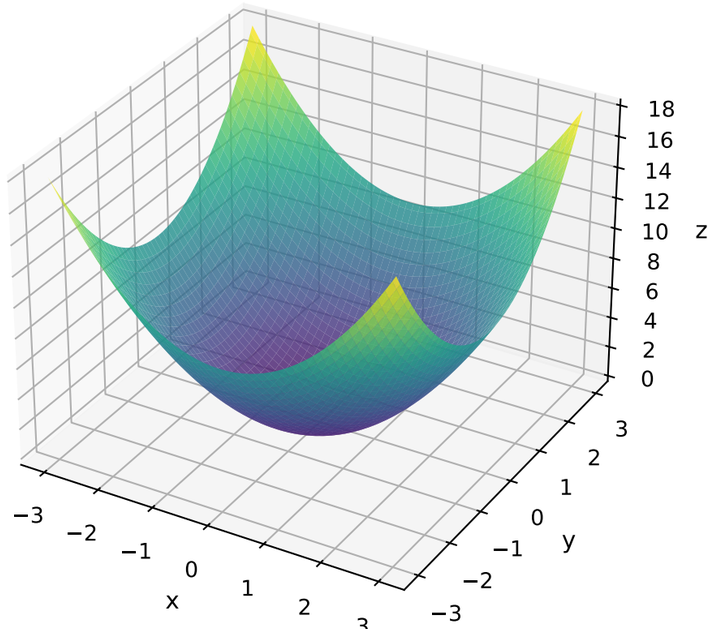



Zadanie 19b: Warstwice funkcji $z = e^{-(x^2+y^2)}$
0.30
0.60
0.80
0.90
0.70
0.50
0.40
0.20
x
y
z
0.00
0.15
0.30
0.45
0.60
0.75
0.90

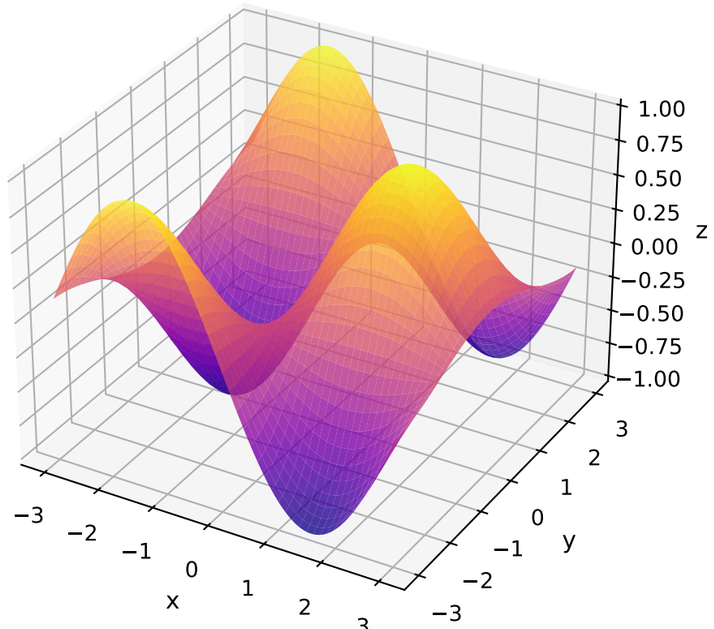

3D: $z = e^{-(x^2+y^2)}$
1.0
0.8
0.6
0.4
0.2
0.0
z
−3
−2
−1
0
1
2
3
x
y


Warstwice
0.2
0.4
0.5
0.7
0.9
0.8
0.6
0.3
y
x
z
0.90
0.75
0.60
0.45
0.30
0.15
0.00

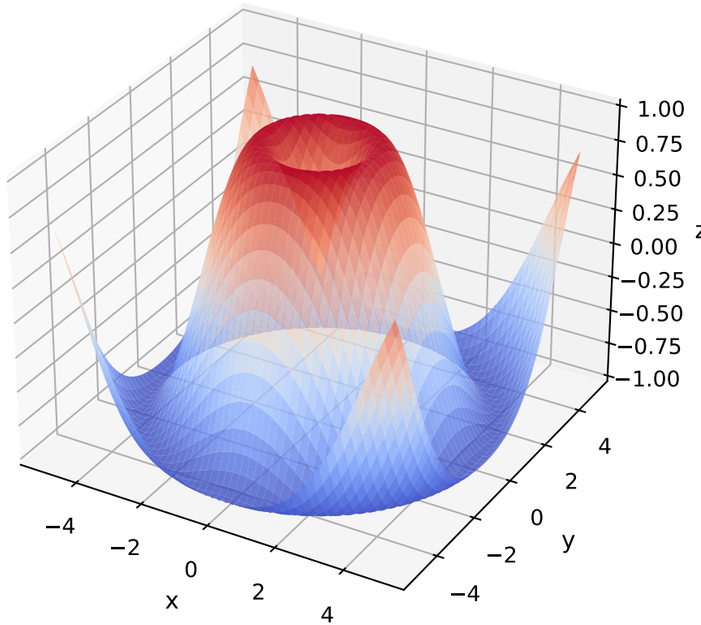
Paraboloida



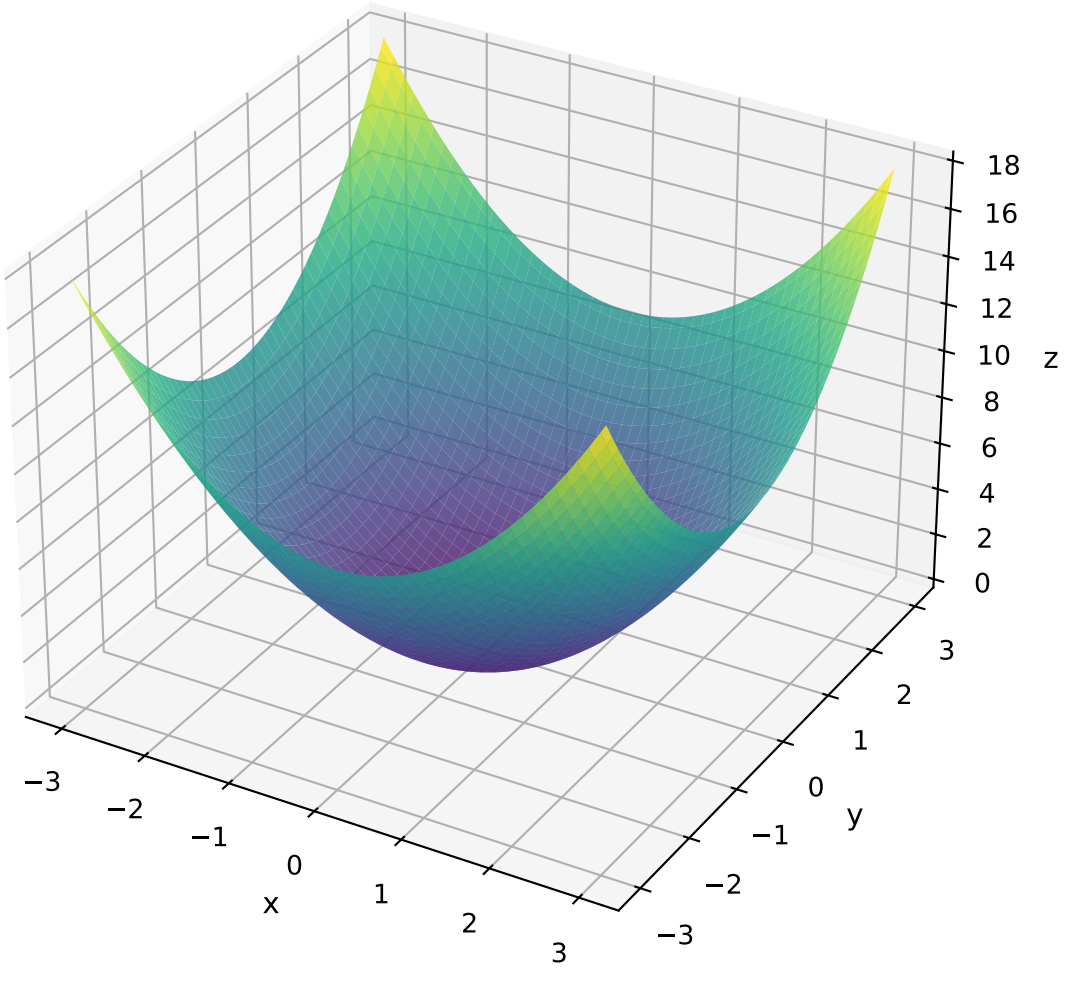
$\sin(x)\cos(y)$



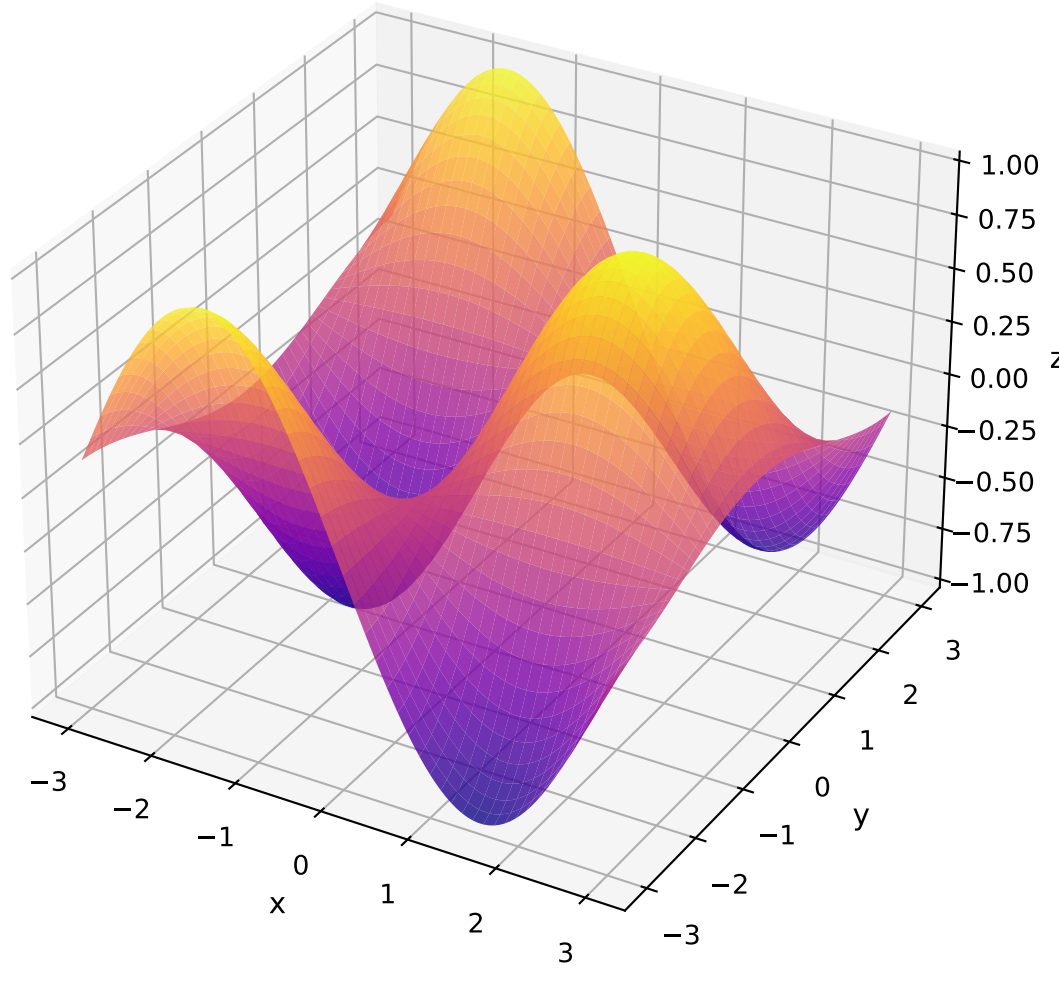
Fale $\sin(\sqrt{x^2+y^2})$



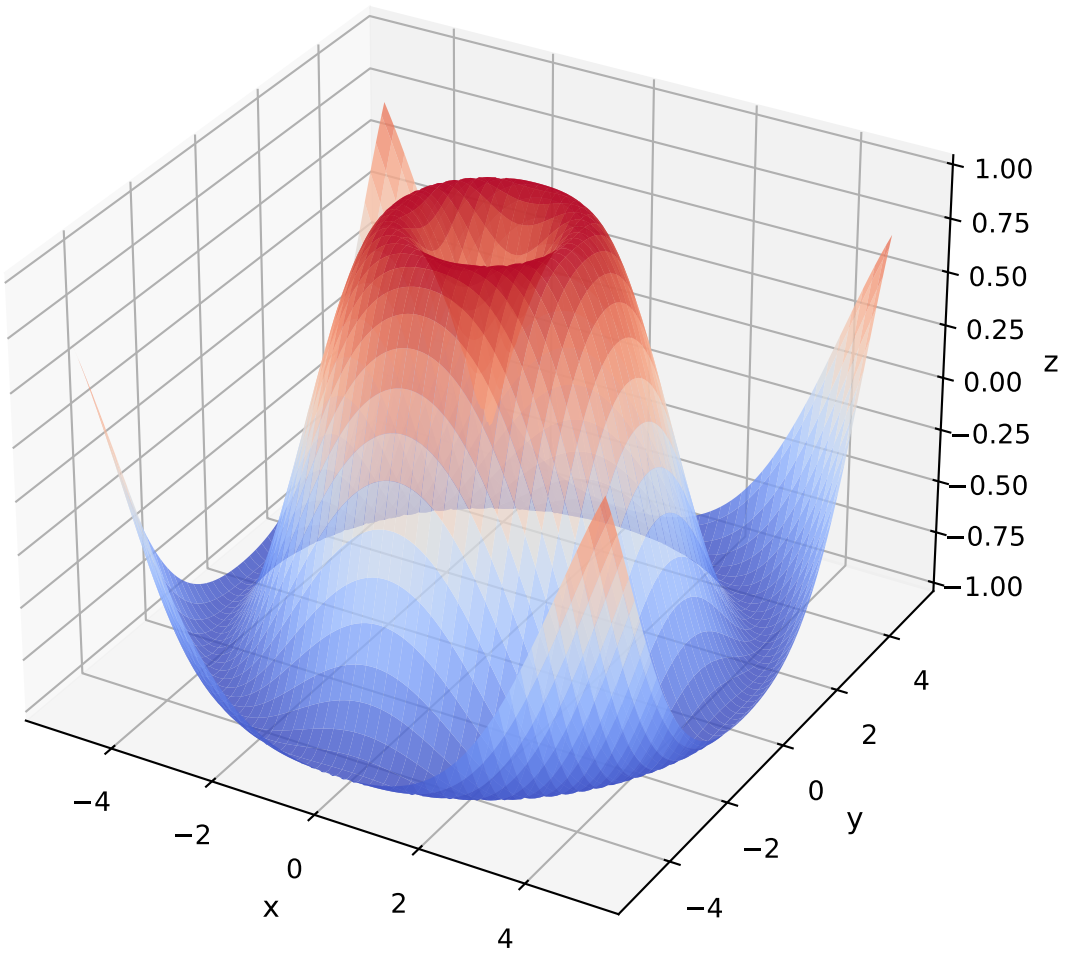
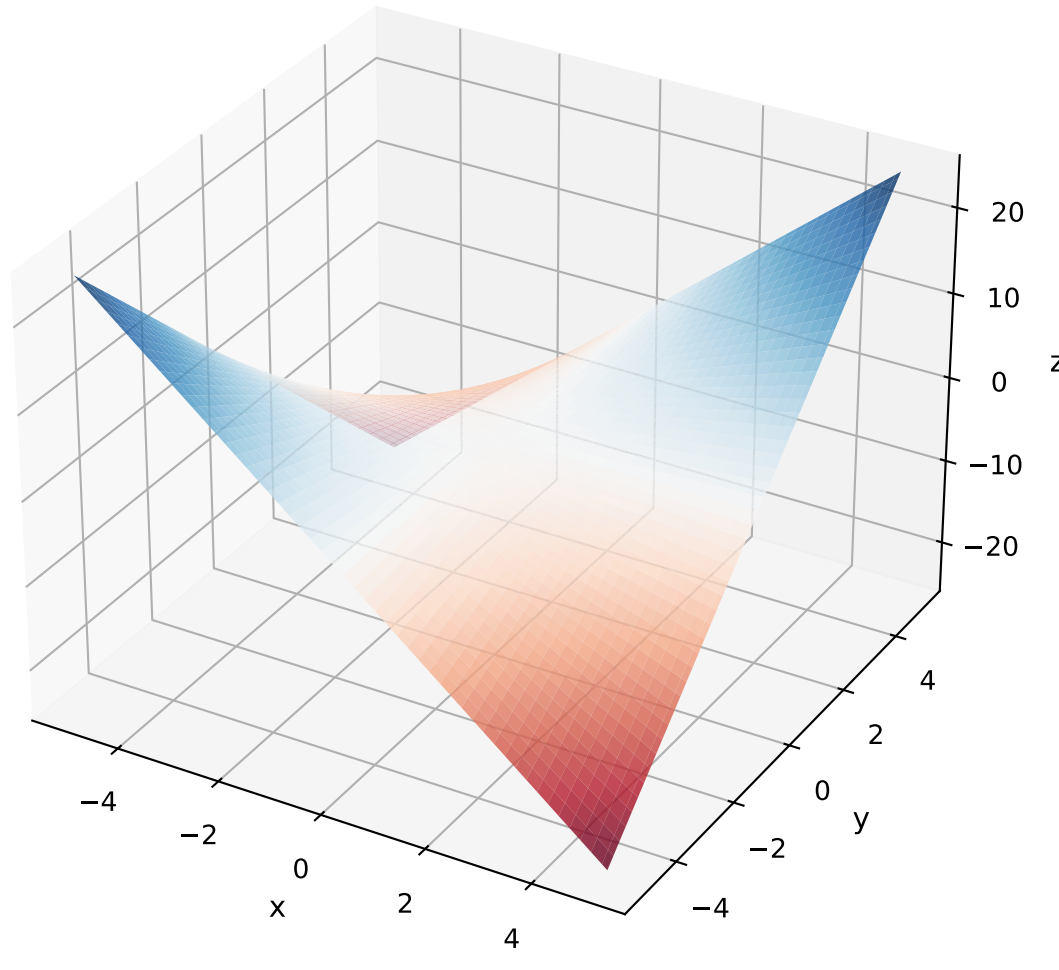
Paraboloida: $z = x^2 + y^2$



$\sin(x)\cos(y)$



Fale: $\sin(\sqrt{x^2 + y^2})$



Siodło: $z = xy$